

HRMS (Human Resource Management System)

Business Requirement Document (BRD)

Technology Stack

Item	Technology
Project Name	Human Resource Management System (HRMS)
Backend	Core Java
Frontend	JSP, Servlet, JSTL
UI Framework	Bootstrap 5
Client Side	HTML5, CSS3, JavaScript
Database	MySQL
Server	Apache Tomcat
IDE	Eclipse IDE/IntelliJ IDE
Architecture	MVC (Model View Controller)
Version Control	Git

1. Introduction

1.1 Purpose

The Human Resource Management System (HRMS) is a web-based application designed to automate and simplify all Human Resource activities within an organization. The system manages employee information, attendance, leave, payroll, projects, training, events, documents, promotions, resignations, terminations, help desk tickets, and various management reports.

The objective of this application is to eliminate manual paperwork, improve operational efficiency, ensure secure employee data management, and provide real-time access to HR-related information.

The application follows the MVC architecture using Core Java, JSP, Servlet, JSTL, Bootstrap, HTML, CSS, JavaScript, and MySQL.

1.2 Business Objective

The primary objectives of the HRMS application are:

- Automate all HR operations.
- Maintain centralized employee records.
- Manage attendance and working hours.
- Process employee leave requests.
- Generate monthly payroll automatically.
- Track employee projects and assigned tasks.
- Store employee and company documents digitally.
- Record promotions, resignations, and terminations.
- Provide an internal help desk through ticket management.
- Generate real-time HR reports for management.
- Reduce manual errors.
- Improve employee productivity.
- Ensure secure role-based access.

1.3 Problem Statement

Many organizations still maintain HR records using Excel sheets, paper files, and multiple disconnected systems. This leads to:

- Duplicate employee records.
- Manual attendance calculation.
- Payroll errors.
- Delayed leave approvals.

- Difficulty tracking employee performance.
- Poor document management.
- Lack of centralized reporting.
- Time-consuming HR processes.
- Increased operational costs.

The HRMS application addresses these challenges by providing a centralized web-based platform.

1.4 Scope of the Project

The HRMS application covers all major Human Resource operations including:

- Employee Management
- Department Management
- Designation Management
- Role Management
- Event Management
- Holiday Calendar
- Trainer Management
- Training Management
- Attendance Management
- Timesheet Management
- Leave Management
- Payroll Management
- Salary Slip Generation
- Project Management
- Task Management
- Task Board
- Document Management

- Promotion Management
- Resignation Management
- Termination Management
- Raise Ticket Module
- Dashboard
- Reports
- User Authentication
- Role-Based Authorization

• 1.5 Stakeholders

Stakeholder	Responsibility
HR Manager	Employee Management, Payroll, Leave, Reports
HR Executive	Daily HR Operations
Administrator	Complete System Management
Employee	Attendance, Leave, Timesheet, Documents
Team Manager	Leave Approval, Project Monitoring
Finance Department	Payroll Processing
Trainer	Training Management
Management	Reports and Analytics

1.6 User Roles

The application supports multiple user roles.

Administrator

Administrator has complete control over the system.

Responsibilities include:

- Employee Management
- Role Management
- Department Management
- Payroll Processing

- Reports
- Training
- Events
- Projects
- Documents
- Promotions
- Resignation Approval
- Termination
- Ticket Resolution

HR Manager

Responsibilities include:

- Employee Management
- Attendance Review
- Payroll Processing
- Leave Approval
- Promotion
- Reports
- Documents

Manager

Responsibilities include:

- Approve Leave
- Assign Tasks
- Track Employee Performance
- Project Monitoring
- Timesheet Approval

Employee

Responsibilities include:

- View Dashboard
- Check Attendance
- Apply Leave
- Fill Timesheet
- Upload Documents
- Raise Tickets
- View Salary Slip
- View Holiday Calendar

1.7 Assumptions

- Every employee has a unique Employee ID.
- One employee belongs to one department.
- One employee has one designation.
- One employee can work on multiple projects.
- Attendance is recorded daily.
- Payroll is generated monthly.
- Leave allocation depends on department policy.
- Employees can upload personal documents.
- Managers approve employee leave.
- Salary slips are generated automatically every month.

1.8 Constraints

- Only authenticated users can access the application.
- Employees cannot approve their own leave.
- Salary cannot be modified after payroll generation.
- Attendance cannot be edited after payroll processing without authorization.
- Deleted employees will remain in history for reporting purposes.

- All uploaded documents must follow allowed file types.
- Role-based security is mandatory.

1.9 Features Summary

The HRMS application consists of the following major modules:

1. Employee Management
2. Event Management
3. Training Management
4. Attendance Management
5. Leave Management
6. Payroll Management
7. Project Management
8. Document Management
9. Promotion Management
10. Resignation Management
11. Termination Management
12. Raise Ticket Management
13. Reports
14. Dashboard
15. Authentication & Authorization

2. Functional Requirements

2.1 Overview

The HRMS application consists of multiple functional modules that automate Human Resource activities. Each module provides Create, Read, Update, Delete (CRUD) operations, business validations, approval workflows, and reporting features based on user roles.

Module 1 – Employee Management

Purpose

The Employee Management module is the core module of the HRMS application. It stores and manages complete employee information from joining to resignation or termination.

It allows HR and administrators to create employee records, assign departments, roles, designations, maintain employment history, and manage employee status.

Users

Role	Permission
Administrator	Full Access
HR Manager	Full Access
HR Executive	Create, Update, View
Manager	View Team Employees
Employee	View Own Profile

Features

The Employee Management module consists of the following sub-modules:

- Department Management
- Designation Management
- Role Management
- Employee Management
- Employee Profile
- Employee Login

- Employee Status
- Employee Search
- Employee Filter
- Employee Import
- Employee Export

Employee Workflow

Create Department

↓

Create Designation

↓

Create Role

↓

Create Employee

↓

Generate Employee ID

↓

Create Login Credentials

↓

Assign Department

↓

Assign Designation

↓

Assign Role

↓

Employee Active

Functional Requirements

FR-EMP-001

Department Management

The system shall allow the Administrator to manage company departments.

Functions

- Add Department
- Update Department
- Delete Department
- Activate Department
- Deactivate Department
- Search Department
- View Department List

Department Details

Field	Type	Required
Department ID	Auto	Yes
Department Name	Text	Yes
Department Code	Text	Yes
Description	TextArea	No
Status	Active/Inactive	Yes
Created Date	DateTime	Auto

Validations

- Department Name cannot be blank.
 - Department Name must be unique.
 - Department Code must be unique.
 - Deleted department cannot be assigned to employees.
-

Business Rules

- One department can have multiple employees.
- One employee belongs to only one department.

FR-EMP-002

Designation Management

The system shall allow HR to maintain employee designations.

Functions

- Add Designation
- Update Designation
- Delete Designation
- Search
- View List

Designation Details

Field	Type
Designation ID	Auto
Department	Dropdown
Designation Name	Text
Description	TextArea
Status	Active/Inactive

Validations

Designation Name is mandatory.

Duplicate Designation is not allowed within the same department.

Business Rules

One Department



Many Designations

Example

IT Department



Software Engineer



Senior Software Engineer



Team Lead



Project Manager

FR-EMP-003

Role Management

Roles define user permissions inside the HRMS application.

Functions

- Add Role
- Update Role
- Delete Role
- Assign Permissions
- View Roles

Roles

Administrator

HR Manager

HR Executive

Manager

Employee

Trainer

Finance

Field	Type
Role ID	Auto
Role Name	Text
Description	TextArea
Status	Active

Business Rule

One Role

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Multiple Users

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FR-EMP-004

Employee Management

The Employee module stores complete employee information.

Functions

- Add Employee
 - Edit Employee
 - Delete Employee
 - View Employee
 - Search Employee
 - Filter Employee
 - Export Excel
 - Export PDF
 - Employee Profile
-

Employee Information

Personal Information

- Employee ID
- First Name
- Middle Name
- Last Name
- Gender
- DOB
- Blood Group
- Marital Status
- Mobile Number
- Alternate Number
- Email
- Aadhaar Number
- PAN Number
- Passport Number
- Nationality

- Profile Photo

Address Information

- Permanent Address
- Current Address
- City
- State
- Country
- PIN Code

Employment Information

- Joining Date
- Employee Type
- Department
- Designation
- Reporting Manager
- Role
- Employment Status
- Work Location
- Shift

Salary Information

- Basic Salary
- HRA
- Allowances
- PF Number
- UAN Number
- Bank Name
- IFSC Code

- Account Number

Emergency Contact

- Contact Person
- Relation
- Mobile Number

Education Details

- Degree
 - University
 - Passing Year
 - Percentage
-

Experience Details

- Company Name
 - Experience
 - Previous Salary
 - Designation
-

Employee Status

Active

Inactive

Notice Period

Resigned

Terminated

Retired

On Leave

Employee Search Filters

- Employee ID
- Employee Name
- Department
- Designation
- Role
- Joining Date
- Status
- Email
- Mobile Number

Validations

- Employee ID must be unique.
- Email must be unique.
- Aadhaar Number must be unique.
- PAN Number must be unique.
- Mobile Number must contain 10 digits.
- Joining Date cannot be a future date.
- DOB cannot be greater than today's date.
- Salary cannot be negative.

Business Rules

- One Employee belongs to one Department.
- One Employee has one Designation.
- One Employee has one Role.
- One Employee reports to one Manager.
- Employee cannot login without Active status.

- Deleted employees are marked Inactive instead of permanently deleted.

Dashboard Information

The Employee Dashboard displays:

- Total Employees
- Active Employees
- Inactive Employees
- New Joiners
- Employees on Leave
- Upcoming Birthdays
- Work Anniversary
- Department-wise Employee Count
- Gender Ratio
- Employee Distribution Chart

Reports

- Employee Master Report
- Department-wise Employee Report
- Designation Report
- Active Employee Report
- Inactive Employee Report
- Joining Report
- Birthday Report
- Anniversary Report
- Experience Report

3. Database Design (Employee Module)

3.1 Overview

The Employee Module is the foundation of the Human Resource Management System (HRMS). It maintains all employee-related master data, personal information, employment details, organizational hierarchy, banking information, educational qualifications, work experience, emergency contacts, and login credentials.

The database is normalized to reduce redundancy, maintain data integrity, and establish relationships between different entities using Primary Keys (PK) and Foreign Keys (FK).

Database Tables

The Employee Module consists of the following tables:

Table No	Table Name	Purpose
EMP001	department	Stores department details
EMP002	designation	Stores designation details
EMP003	role	Stores user roles
EMP004	employee	Stores employee master information
EMP005	employee_login	Stores login credentials
EMP006	employee_address	Stores permanent/current addresses
EMP007	employee_bank	Stores bank details
EMP008	employee_education	Stores educational qualifications
EMP009	employee_experience	Stores previous employment details
EMP010	employee_emergency_contact	Stores emergency contact details
EMP011	employee_documents	Stores employee uploaded documents

Table 3.1 : department

Purpose

Stores all company departments.

Column Name	Data Type	Size	Constraint	Description
department_id	INT	-	PK, Auto Increment	Department ID
department_name	VARCHAR	100	NOT NULL, UNIQUE	Department Name
department_code	VARCHAR	20	UNIQUE	Department Code
description	VARCHAR	255	NULL	Department Description
status	ENUM('Active','Inactive')	-	Default Active	Department Status
created_by	INT	-	FK	Created User
created_date	DATETIME	-	Default CURRENT_TIMESTAMP	Created Date
updated_by	INT	-	FK	Updated User
updated_date	DATETIME	-	NULL	Updated Date

Business Rules

- Department Name must be unique.
- Department Code must be unique.
- Inactive departments cannot be assigned to new employees.
- Department cannot be deleted if employees are assigned.

Table 3.2 : designation

Purpose

Stores employee designations.

Column Name	Data Type	Size	Constraint
designation_id	INT	-	PK, Auto Increment
department_id	INT	-	FK
designation_name	VARCHAR	100	NOT NULL
description	VARCHAR	255	NULL
status	ENUM('Active','Inactive')	-	Default Active
created_date	DATETIME	-	Current Timestamp

Table 3.3 : role

Purpose

Stores application roles.

Column Name	Data Type	Constraint
role_id	INT	PK
role_name	VARCHAR(50)	UNIQUE
description	VARCHAR(255)	NULL
status	ENUM('Active','Inactive')	Default Active
created_date	DATETIME	Current Timestamp

Roles

- Administrator
- HR Manager
- HR Executive
- Team Manager
- Employee
- Finance
- Trainer

Table 3.4 : employee

Purpose

Stores complete employee master information.

Column Name	Data Type	Constraint
employee_id	INT	PK Auto Increment
employee_code	VARCHAR(20)	UNIQUE
first_name	VARCHAR(50)	NOT NULL
middle_name	VARCHAR(50)	NULL
last_name	VARCHAR(50)	NOT NULL
gender	ENUM('Male','Female','Other')	NOT NULL
date_of_birth	DATE	NOT NULL
blood_group	VARCHAR(10)	NULL
marital_status	ENUM('Single','Married')	NULL
mobile	VARCHAR(15)	UNIQUE

Column Name	Data Type	Constraint
alternate_mobile	VARCHAR(15)	NULL
email	VARCHAR(100)	UNIQUE
aadhaar_number	VARCHAR(12)	UNIQUE
pan_number	VARCHAR(10)	UNIQUE
passport_number	VARCHAR(20)	NULL
nationality	VARCHAR(50)	NULL
profile_photo	VARCHAR(255)	NULL
joining_date	DATE	NOT NULL
department_id	INT	FK
designation_id	INT	FK
role_id	INT	FK
manager_id	INT	FK (employee_id)
shift_id	INT	FK
employment_type	ENUM('Permanent','Contract','Intern')	NOT NULL
employment_status	ENUM('Active','Inactive','Resigned','Terminated','Notice')	Default Active
created_date	DATETIME	Current Timestamp

Business Rules

- Employee Code is auto-generated.

- Email must be unique.
- Aadhaar Number must be unique.
- PAN Number must be unique.
- Mobile Number must be unique.
- One employee belongs to one department.
- One employee has one designation.
- One employee reports to one manager.

Table 3.5 : employee_login

Purpose

Stores employee login credentials.

Column Name	Data Type	Constraint
login_id	INT	PK
employee_id	INT	FK
username	VARCHAR(100)	UNIQUE
password	VARCHAR(255)	NOT NULL (Encrypted)
last_login	DATETIME	NULL
login_status	ENUM('Active','Blocked')	Default Active
failed_attempt	INT	Default 0
password_changed_date	DATE	NULL

Business Rules

- Password should be encrypted.
- Maximum 5 failed login attempts.

- After 5 failed attempts account is blocked.
- Only Active employees can login.

Table 3.6 : employee_address

Column Name	Data Type
address_id	INT PK
employee_id	INT FK
address_type	Permanent / Current
address_line1	VARCHAR(255)
address_line2	VARCHAR(255)
city	VARCHAR(100)
state	VARCHAR(100)
country	VARCHAR(100)
pin_code	VARCHAR(10)

Table 3.7 : employee_bank

Column Name	Data Type
bank_id	INT PK
employee_id	INT FK
bank_name	VARCHAR(100)
account_number	VARCHAR(30)
ifsc_code	VARCHAR(20)

Column Name	Data Type
branch_name	VARCHAR(100)
account_type	Savings / Current
uan_number	VARCHAR(20)
pf_number	VARCHAR(30)

Business Rules

- One active bank account per employee for salary credit.
- Account Number should be unique.
- IFSC Code is mandatory.

Table 3.8 : employee_education

Column Name	Data Type
education_id	INT PK
employee_id	INT FK
qualification	VARCHAR(100)
university	VARCHAR(150)
institute	VARCHAR(150)
passing_year	YEAR
percentage	DECIMAL(5,2)
grade	VARCHAR(10)

Business Rules

- One employee can have multiple education records.
- Passing Year cannot be greater than the current year.

Table 3.9 : employee_experience

Column Name	Data Type
experience_id	INT PK
employee_id	INT FK
company_name	VARCHAR(150)
designation	VARCHAR(100)
joining_date	DATE
relieving_date	DATE
experience_years	DECIMAL(4,1)
last_salary	DECIMAL(10,2)
reason_for_leaving	VARCHAR(255)

Business Rules

- One employee can have multiple experience records.
- Relieving Date must be greater than Joining Date.

Table 3.10 : employee_emergency_contact

Column Name	Data Type
contact_id	INT PK

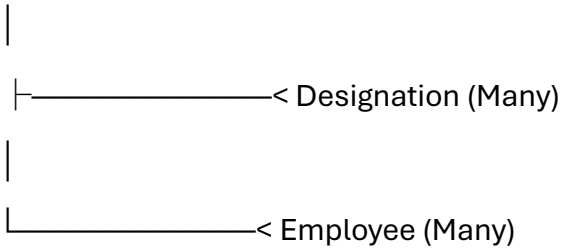
Column Name	Data Type
employee_id	INT FK
contact_name	VARCHAR(100)
relationship	VARCHAR(50)
mobile_number	VARCHAR(15)
alternate_number	VARCHAR(15)
address	VARCHAR(255)

Table 3.11 : employee_documents

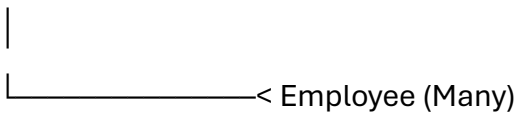
Column Name	Data Type
document_id	INT PK
employee_id	INT FK
document_type	VARCHAR(100)
file_name	VARCHAR(255)
file_path	VARCHAR(255)
file_size	BIGINT
uploaded_date	DATETIME
verification_status	ENUM('Pending','Verified','Rejected')

Entity Relationships

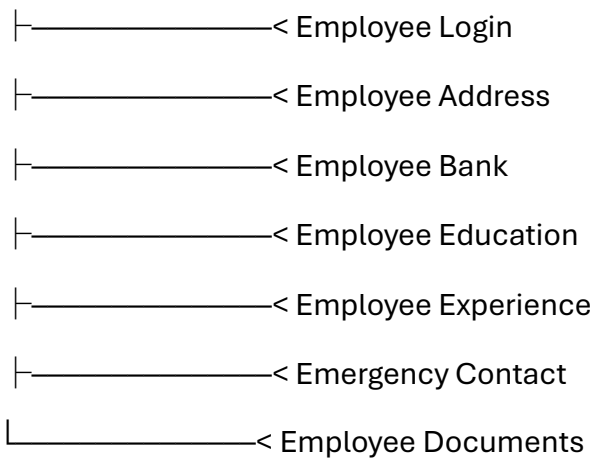
Department (1)



Role (1)



Employee (1)



Deliverables of Employee Module

- Department Master
- Designation Master
- Role Master
- Employee Master
- Employee Profile
- Employee Login

- Employee Address
- Employee Bank Details
- Employee Education
- Employee Experience
- Employee Emergency Contact
- Employee Document Management
- Employee Search & Filters
- Employee Reports

4. Event Management Module

4.1 Module Overview

The Event Management Module is designed to manage all organizational events and holidays within the HRMS application. It enables HR and Administrators to create event categories, schedule company events, maintain holiday calendars, and publish important announcements for employees.

This module provides a centralized calendar where employees can view upcoming birthdays, festivals, meetings, and training schedules.

4.2 Module Objectives

The primary objectives of the Event Management Module are:

- Manage company events.
- Maintain yearly holiday calendar.
- Publish organization announcements.
- Notify employees about upcoming events.
- Avoid scheduling conflicts.
- Improve employee engagement.

- Provide calendar-based event visualization.
- Generate holiday and event reports.

4.3 User Roles

User Role	Permission
Administrator	Full Access
HR Manager	Full Access
HR Executive	Create, Update, View
Manager	View
Employee	View Only

4.4 Module Components

The Event Management Module consists of the following sub-modules:

1. Event Type Master
2. Event Management
3. Holiday Calendar
4. Calendar View
5. Event Notifications
6. Event Reports

4.5 Event Management Workflow

Create Event Type



Create Event



Assign Event Date



Publish Event



Employee Notification



Display in Calendar

Functional Requirements

FR-EVT-001

Event Type Master

Description

The Event Type Master is used to define different categories of company events.

Functions

- Add Event Type
- Edit Event Type
- Delete Event Type
- Activate Event Type
- Deactivate Event Type
- Search Event Type
- View Event Type List

Fields

Field Name	Type	Required
Event Type ID	Auto	Yes
Event Type Name	Text	Yes
Description	TextArea	No
Color Code	Color Picker	Yes
Status	Active/Inactive	Yes
Created Date	DateTime	Auto

Sample Event Types

- Festival
- Birthday
- Meeting
- Seminar
- Workshop

- Training
 - Company Event
 - Sports
 - Cultural Program
 - Holiday
-

Validations

- Event Type Name cannot be blank.
 - Event Type Name must be unique.
 - Color Code is mandatory.
 - Deleted Event Types cannot be assigned to events.
-

Business Rules

- One Event Type can have multiple events.
 - Event Type Name must be unique.
-

FR-EVT-002

Event Management

Description

This module manages all company events.

Functions

- Add Event
- Edit Event
- Delete Event

- Publish Event
- Cancel Event
- Search Event
- Filter Event
- View Calendar

Event Details

Field	Type
Event ID	Auto
Event Type	Dropdown
Event Title	Text
Description	TextArea
Event Date	Date
Start Time	Time
End Time	Time
Venue	Text
Organizer	Dropdown(Employee)
Department	Dropdown
Maximum Participants	Number
Banner Image	File Upload
Status	Draft / Published / Cancelled

Validations

- Event Title is mandatory.
 - Event Date cannot be in the past (except historical entries).
 - End Time must be greater than Start Time.
 - Organizer must be an active employee.
-

Business Rules

- Published events are visible to employees.
 - Cancelled events remain in history.
 - Employees cannot edit events.
-

FR-EVT-003

Holiday Calendar

Description

The Holiday Calendar stores all company holidays for a particular financial or calendar year.

Functions

- Add Holiday
- Edit Holiday
- Delete Holiday
- Import Holidays
- Export Holidays
- View Calendar

Holiday Details

Field	Type
Holiday ID	Auto
Holiday Name	Text
Holiday Date	Date
Holiday Type	National / Festival / Company
Applicable Department	Dropdown
Description	TextArea
Is Optional Holiday	Yes / No
Status	Active

Holiday Types

- National Holiday
- Festival Holiday
- Company Holiday
- Optional Holiday

Validations

- Holiday Name is mandatory.
- Duplicate Holiday Date is not allowed for the same department.
- Holiday Date must be unique.

Business Rules

- Holidays are displayed on Attendance Calendar.

- Holidays are excluded from leave calculations.
- Payroll considers holidays while processing attendance.

FR-EVT-004

Calendar View

Description

The Calendar provides a monthly, weekly, and daily view of company activities.

Calendar Displays

- Company Holidays
 - Events
 - Employee Birthdays
 - Work Anniversaries
 - Training Sessions
 - Leave Calendar
 - Meetings
-

Calendar Filters

- Department
- Event Type
- Employee
- Month
- Year

FR-EVT-005**Event Notification**

Description

The system automatically notifies employees regarding upcoming events.

Notification Channels

- Dashboard Notification
 - Email Notification
 - System Notification
-

Notification Trigger

- New Event Published
 - Event Updated
 - Event Cancelled
 - Holiday Added
-

Reports

The Event Module provides the following reports:

- Event Master Report
- Event Type Report
- Upcoming Events Report
- Holiday Report
- Monthly Event Report
- Department-wise Event Report

- Festival Report
- Birthday Report
- Anniversary Report
- Calendar Report

Database Design

Table EVT001 : event_type

Purpose

Stores event categories.

Column Name	Data Type	Constraint
event_type_id	INT	PK, Auto Increment
event_type_name	VARCHAR(100)	NOT NULL, UNIQUE
description	VARCHAR(255)	NULL
color_code	VARCHAR(20)	NOT NULL
status	ENUM('Active','Inactive')	Default Active
created_by	INT	FK(Employee)
created_date	DATETIME	CURRENT_TIMESTAMP

Table EVT002 : event

Purpose

Stores all company events.

Column Name	Data Type	Constraint
event_id	INT	PK
event_type_id	INT	FK
event_title	VARCHAR(150)	NOT NULL
description	TEXT	NULL
event_date	DATE	NOT NULL
start_time	TIME	NOT NULL
end_time	TIME	NOT NULL
venue	VARCHAR(150)	NULL
organizer_id	INT	FK(Employee)
department_id	INT	FK(Department)
max_participants	INT	Default 0
banner_image	VARCHAR(255)	NULL
status	ENUM('Draft','Published','Cancelled')	Default Draft
created_by	INT	FK(Employee)
created_date	DATETIME	CURRENT_TIMESTAMP

Table EVT003 : holiday

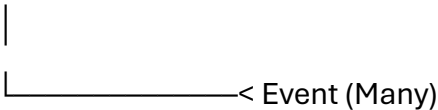
Purpose

Stores all company holidays.

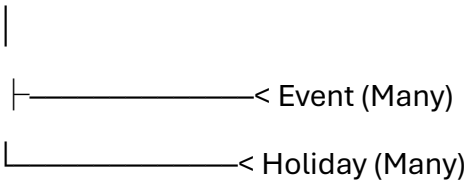
Column Name	Data Type	Constraint
holiday_id	INT	PK
holiday_name	VARCHAR(150)	NOT NULL
holiday_date	DATE	NOT NULL
holiday_type	ENUM('National','Festival','Company','Optional')	NOT NULL
department_id	INT	FK
description	VARCHAR(255)	NULL
is_optional	BOOLEAN	Default FALSE
status	ENUM('Active','Inactive')	Default Active
created_by	INT	FK(Employee)
created_date	DATETIME	CURRENT_TIMESTAMP

Entity Relationship

Event Type (1)



Department (1)



Employee (1)

└──────────> Event (Organizer)

Dashboard Widgets

- Today's Events
- Upcoming Events
- Upcoming Holidays
- Current Month Calendar
- Employee Birthdays
- Work Anniversaries
- Recent Announcements

Deliverables

- Event Type Master
- Event Management
- Holiday Management
- Calendar View
- Notifications
- Event Reports
- Holiday Reports
- Calendar Dashboard

5. Training Management Module

5.1 Module Overview

The Training Management Module is designed to manage employee learning and development activities within the organization. It enables HR and Administrators to create trainers, define training categories, schedule training sessions, assign employees, record attendance, collect feedback, and maintain training history.

This module helps organizations improve employee skills, monitor training effectiveness, and maintain a complete record of all internal and external training programs.

5.2 Module Objectives

The Training Management Module aims to:

- Manage internal and external trainers.
 - Create different training categories.
 - Schedule training programs.
 - Nominate employees for training.
 - Track training attendance.
 - Collect employee feedback.
 - Store training completion certificates.
 - Generate training reports.
 - Maintain employee training history.
-

5.3 User Roles

User Role	Permission
Administrator	Full Access
HR Manager	Full Access

User Role	Permission
HR Executive	Create, Update, View
Trainer	Conduct Training, Mark Attendance
Manager	Nominate Employees, View Reports
Employee	View Assigned Trainings, Submit Feedback

5.4 Module Components

The Training Management Module consists of the following sub-modules:

1. Trainer Master
2. Training Type Master
3. Training Management
4. Employee Training Assignment
5. Training Attendance
6. Training Feedback
7. Training Certificate Management
8. Training Reports

5.5 Training Workflow

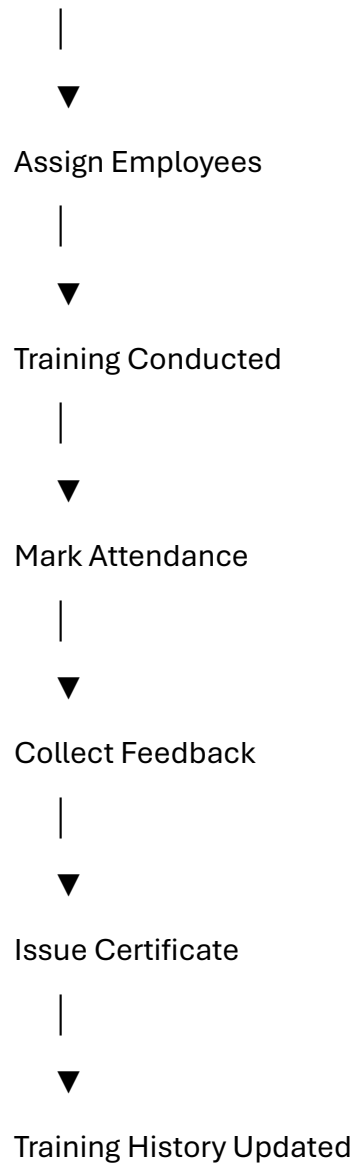
Create Trainer



Create Training Type



Schedule Training



Functional Requirements

FR-TRN-001

Trainer Master

Description

The Trainer Master stores details of all internal and external trainers who conduct training programs.

Functions

- Add Trainer
 - Edit Trainer
 - Delete Trainer
 - Search Trainer
 - View Trainer List
 - Activate / Deactivate Trainer
-

Trainer Details

Field	Type
Trainer ID	Auto
Trainer Name	Text
Trainer Type	Internal / External
Employee	Dropdown (Internal Trainer)
Company Name	Text (External Trainer)
Mobile Number	Text
Email	Email
Specialization	Text
Experience (Years)	Number
Qualification	Text
Status	Active / Inactive

Validations

- Trainer Name is mandatory.
 - Email must be unique.
 - Mobile Number must contain 10 digits.
 - Internal Trainer must be an active employee.
-

Business Rules

- One trainer can conduct multiple training sessions.
 - Inactive trainers cannot be assigned to new training.
-

FR-TRN-002

Training Type Master

Description

Training Type Master categorizes different types of organizational training.

Functions

- Add Training Type
 - Edit Training Type
 - Delete Training Type
 - Search
 - View List
-

Fields

Field	Type
Training Type ID	Auto

Field	Type
Training Type Name	Text
Description	TextArea
Duration (Hours)	Number
Status	Active / Inactive

Sample Training Types

- Technical Training
- Soft Skills
- Product Training
- Induction
- Compliance
- Leadership
- Safety
- Communication
- Customer Service

Validations

- Training Type Name must be unique.
- Duration must be greater than zero.

FR-TRN-003

Training Management

Description

This module schedules and manages all training programs.

Functions

- Create Training
 - Update Training
 - Cancel Training
 - Publish Training
 - View Calendar
 - Search Training
-

Training Details

Field	Type
Training ID	Auto
Training Name	Text
Training Type	Dropdown
Trainer	Dropdown
Department	Dropdown
Venue	Text
Start Date	Date
End Date	Date
Start Time	Time
End Time	Time
Maximum Participants	Number
Training Cost	Decimal

Field	Type
Status	Draft / Published / Completed / Cancelled

Business Rules

- End Date must be greater than Start Date.
- Maximum Participants cannot exceed room capacity.
- Cancelled training remains in history.
- Published training is visible to employees.

FR-TRN-004

Employee Training Assignment

Description

Assign employees to scheduled training sessions.

Functions

- Assign Employee
- Remove Employee
- View Assigned Employees
- Search Employees

Assignment Details

Field	Type
Assignment ID	Auto
Training	Dropdown

Field	Type
Employee	Dropdown
Department	Auto
Assigned Date	Date
Status	Assigned / Completed / Cancelled

Business Rules

- Employee cannot be assigned twice to the same training.
 - Only active employees can be assigned.
-

FR-TRN-005

Training Attendance

Description

Record attendance of employees during training sessions.

Attendance Status

- Present
 - Absent
 - Late
 - Half Day
-

Functions

- Mark Attendance
- Edit Attendance
- View Attendance

- Export Attendance

Fields

Field	Type
Attendance ID	Auto
Training	Dropdown
Employee	Dropdown
Attendance Date	Date
Status	Present / Absent / Late / Half Day
Remarks	TextArea

FR-TRN-006**Training Feedback****Description**

Collect employee feedback after completion of training.

Feedback Details

Field	Type
Feedback ID	Auto
Training	Dropdown
Employee	Dropdown
Trainer Rating	1–5

Field	Type
Content Rating	1–5
Overall Rating	1–5
Comments	TextArea
Feedback Date	Date

Business Rules

- Feedback can be submitted only once.
- Feedback is allowed only after training completion.

FR-TRN-007

Training Certificate

Description

Manage certificates issued after successful completion of training.

Certificate Details

Field	Type
Certificate ID	Auto
Training	Dropdown
Employee	Dropdown

Field	Type
Certificate Number	Auto
Issue Date	Date
Certificate File	Upload
Status	Issued / Pending

Reports

The Training Module provides the following reports:

- Trainer Master Report
- Training Type Report
- Training Schedule Report
- Department-wise Training Report
- Employee Training History Report
- Training Attendance Report
- Training Completion Report
- Training Feedback Report
- Certificate Report

Database Design

Table TRN001 : trainer

Column Name	Data Type	Constraint
trainer_id	INT	PK, Auto Increment
trainer_name	VARCHAR(100)	NOT NULL
trainer_type	ENUM('Internal','External')	NOT NULL
employee_id	INT	FK(Employee), NULL
company_name	VARCHAR(150)	NULL
mobile_number	VARCHAR(15)	NULL
email	VARCHAR(100)	UNIQUE
specialization	VARCHAR(150)	NULL
experience_years	DECIMAL(4,1)	NULL
qualification	VARCHAR(100)	NULL
status	ENUM('Active','Inactive')	Default Active
created_date	DATETIME	CURRENT_TIMESTAMP

Table TRN002 : training_type

Column Name	Data Type	Constraint
training_type_id	INT	PK
training_type_name	VARCHAR(100)	UNIQUE
description	VARCHAR(255)	NULL
duration_hours	INT	NOT NULL
status	ENUM('Active','Inactive')	Default Active

Table TRN003 : training

Column Name	Data Type	Constraint
training_id	INT	PK
training_name	VARCHAR(150)	NOT NULL
training_type_id	INT	FK
trainer_id	INT	FK
department_id	INT	FK
venue	VARCHAR(150)	NULL
start_date	DATE	NOT NULL
end_date	DATE	NOT NULL
start_time	TIME	NOT NULL
end_time	TIME	NOT NULL
max_participants	INT	Default 0
training_cost	DECIMAL(10,2)	NULL
status	ENUM('Draft','Published','Completed','Cancelled')	Default Draft
created_by	INT	FK(Employee)
created_date	DATETIME	CURRENT_TIMESTAMP

Table TRN004 : training_assignment

Column Name	Data Type	Constraint
assignment_id	INT	PK
training_id	INT	FK
employee_id	INT	FK
assigned_date	DATE	NOT NULL
status	ENUM('Assigned','Completed','Cancelled')	Default Assigned

Table TRN005 : training_attendance

Column Name	Data Type	Constraint
attendance_id	INT	PK
training_id	INT	FK
employee_id	INT	FK
attendance_date	DATE	NOT NULL
attendance_status	ENUM('Present','Absent','Late','Half Day')	NOT NULL
remarks	VARCHAR(255)	NULL

Table TRN006 : training_feedback

Column Name	Data Type	Constraint
feedback_id	INT	PK
training_id	INT	FK
employee_id	INT	FK

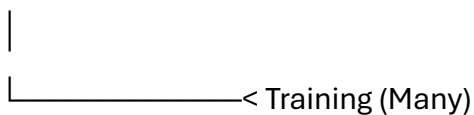
Column Name	Data Type	Constraint
trainer_rating	INT	CHECK (trainer_rating BETWEEN 1 AND 5)
content_rating	INT	CHECK (content_rating BETWEEN 1 AND 5)
overall_rating	INT	CHECK (overall_rating BETWEEN 1 AND 5)
comments	TEXT	NULL
feedback_date	DATE	NOT NULL

Table TRN007 : training_certificate

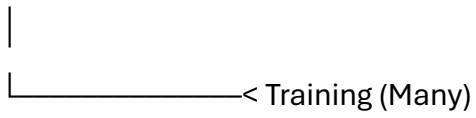
Column Name	Data Type	Constraint
certificate_id	INT	PK
training_id	INT	FK
employee_id	INT	FK
certificate_number	VARCHAR(50)	UNIQUE
issue_date	DATE	NOT NULL
certificate_file	VARCHAR(255)	NULL
status	ENUM('Issued','Pending')	Default Pending

Entity Relationship

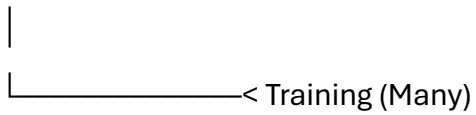
Trainer (1)



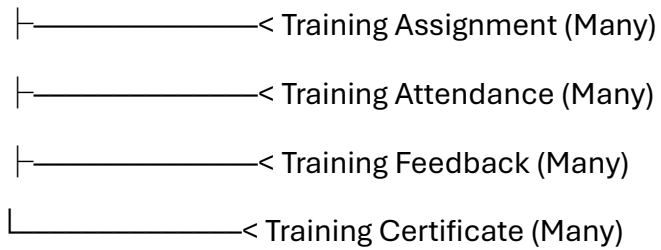
Training Type (1)



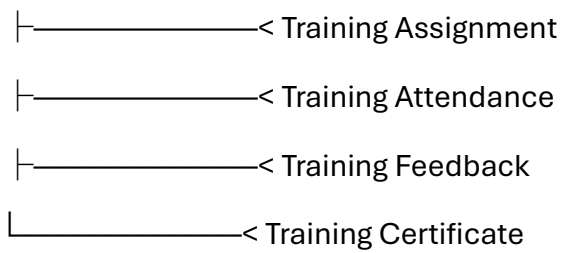
Department (1)



Training (1)



Employee (1)



Dashboard Widgets

- Total Trainings
- Upcoming Trainings
- Completed Trainings
- Active Trainers
- Training Calendar

- Department-wise Training Count
 - Employee Training Completion %
 - Pending Feedback
 - Certificates Issued
-

Deliverables

- Trainer Master
 - Training Type Master
 - Training Scheduling
 - Employee Training Assignment
 - Training Attendance
 - Training Feedback
 - Certificate Management
 - Training Reports
 - Training Dashboard
-

6. Attendance & Timesheet Management Module

6.1 Module Overview

The Attendance & Timesheet Management Module is one of the core modules of the HRMS application. It records employees' daily attendance, working hours, breaks, overtime, late arrivals, early departures, and project-wise timesheets.

This module integrates with Leave Management and Payroll to calculate payable working days, salary deductions, overtime, and monthly attendance reports.

6.2 Module Objectives

The Attendance & Timesheet Module aims to:

- Record employee attendance.
- Track Check-In and Check-Out timings.
- Record Lunch Break timings.
- Calculate total worked hours.
- Calculate Late Arrival and Early Exit.
- Calculate Overtime.
- Maintain employee timesheets.
- Generate monthly attendance reports.
- Integrate attendance with payroll.

6.3 User Roles

User Role	Permission
Administrator	Full Access
HR Manager	Full Access
HR Executive	Manage Attendance
Team Manager	Approve Timesheets
Employee	Mark Attendance & Fill Timesheet

6.4 Module Components

The Attendance Module consists of:

1. Shift Master
2. Attendance Policy
3. Employee Attendance
4. Attendance Regularization

- 5. Overtime Management
- 6. Timesheet Management
- 7. Monthly Attendance Summary
- 8. Attendance Reports

6.5 Attendance Workflow

Employee Login



Check-In



Lunch-In



Lunch-Out



Check-Out



Calculate Worked Hours



Calculate Late / Early Exit





Attendance Summary



Payroll Processing

Functional Requirements

FR-ATT-001

Shift Master

Description

The Shift Master defines official working shifts for employees.

Functions

- Add Shift
 - Update Shift
 - Delete Shift
 - Assign Shift
 - Activate/Deactivate Shift
-

Shift Details

Field	Type
Shift ID	Auto
Shift Name	Text
Shift Start Time	Time

Field	Type
Shift End Time	Time
Break Duration	Number (Minutes)
Grace Time	Number (Minutes)
Working Hours	Decimal
Weekly Off	Multi Select
Status	Active / Inactive

Sample Shifts

- General Shift (09:00 AM – 06:00 PM)
- Morning Shift
- Evening Shift
- Night Shift
- Flexible Shift

Business Rules

- Shift Name must be unique.
- Grace Time is considered while calculating late arrivals.
- One employee can have only one active shift.

FR-ATT-002

Attendance Policy

Description

Attendance Policy defines attendance rules applicable across the organization.

Policy Settings

- Office Start Time
 - Office End Time
 - Grace Time
 - Minimum Working Hours
 - Half-Day Hours
 - Overtime Eligibility
 - Weekend Policy
 - Holiday Policy
 - Late Mark Rules
-

Example Policy

Setting	Value
Office Timing	09:00 AM – 06:00 PM
Working Hours	9 Hours
Lunch Break	1 Hour
Grace Time	15 Minutes
Half Day	Less than 4.5 Hours
Overtime	More than 9 Hours

FR-ATT-003

Employee Attendance

Description

Employees mark their daily attendance using Check-In and Check-Out.

Attendance Flow

Check-In

↓

Lunch-In

↓

Lunch-Out

↓

Check-Out

↓

Attendance Calculation

Attendance Details

Field	Type
Attendance ID	Auto
Employee	Dropdown
Attendance Date	Date
Check-In Time	Time
Lunch-In Time	Time
Lunch-Out Time	Time
Check-Out Time	Time
Worked Hours	Decimal
Overtime Hours	Decimal
Late Minutes	Number
Early Exit Minutes	Number
Attendance Status	Present / Absent / Half Day / Leave / Holiday
Remarks	TextArea

Attendance Status

- Present
- Absent
- Half Day
- Leave
- Holiday
- Weekend
- Work From Home

Business Rules

- Only one attendance record per employee per day.
- Check-Out time must be greater than Check-In time.
- Lunch-Out must be after Lunch-In.
- Attendance cannot be edited after payroll is processed without HR approval.

FR-ATT-004

Attendance Regularization

Description

Employees can request attendance correction if they forgot to check in/out or incorrect attendance is recorded.

Functions

- Apply Regularizaion
- Manager Approval
- HR Approval
- Reject Request

Regularization Details

Field	Type
Request ID	Auto
Employee	Dropdown
Attendance Date	Date

Field	Type
Reason	TextArea
Corrected Check-In	Time
Corrected Check-Out	Time
Status	Pending / Approved / Rejected

Business Rules

- Attendance can be regularized only once per day.
 - Only pending requests can be edited.
 - Manager approval is mandatory before HR approval.
-

FR-ATT-005

Overtime Management

Description

Tracks additional hours worked beyond official shift timings.

Overtime Details

Field	Type
Overtime ID	Auto
Employee	Dropdown
Attendance Date	Date
Worked Hours	Decimal
Overtime Hours	Decimal

Field	Type
Approved By	Manager
Status	Pending / Approved / Rejected

Business Rules

- Overtime is calculated after completion of shift hours.
- Overtime requires manager approval.
- Approved overtime is considered during payroll processing.

FR-ATT-006

Timesheet Management

Description

Employees record daily project-wise work hours.

Functions

- Add Timesheet
- Edit Timesheet
- Delete Timesheet
- Submit Timesheet
- Approve Timesheet

Timesheet Details

Field	Type
Timesheet ID	Auto

Field	Type
Employee	Dropdown
Project	Dropdown
Task	Dropdown
Date	Date
Worked Hours	Decimal
Description	TextArea
Status	Draft / Submitted / Approved / Rejected

Business Rules

- Worked Hours cannot exceed 24 hours in one day.
 - Timesheet can be edited only before submission.
 - Manager approval is mandatory.
 - Total Timesheet Hours should match Attendance Worked Hours (configurable).
-

Reports

The Attendance Module provides:

- Daily Attendance Report
- Monthly Attendance Report
- Employee Attendance Report
- Late Arrival Report
- Early Exit Report
- Overtime Report
- Shift-wise Attendance Report

- Department-wise Attendance Report
- Timesheet Report
- Pending Timesheet Approval Report

Database Design

Table ATT001 : shift_master

Column Name	Data Type	Constraint
shift_id	INT	PK
shift_name	VARCHAR(100)	UNIQUE
start_time	TIME	NOT NULL
end_time	TIME	NOT NULL
break_minutes	INT	DEFAULT 60
grace_minutes	INT	DEFAULT 15
working_hours	DECIMAL(4,2)	NOT NULL
weekly_off	VARCHAR(100)	NOT NULL
status	ENUM('Active','Inactive')	Default Active

Table ATT002 : attendance_policy

Column Name	Data Type	Constraint
policy_id	INT	PK
office_start	TIME	NOT NULL
office_end	TIME	NOT NULL

Column Name	Data Type	Constraint
grace_time	INT	NOT NULL
half_day_hours	DECIMAL(4,2)	NOT NULL
minimum_hours	DECIMAL(4,2)	NOT NULL
overtime_after	DECIMAL(4,2)	NOT NULL
status	ENUM('Active','Inactive')	Default Active

Table ATT003 : employee_attendance

Column Name	Data Type	Constraint
attendance_id	INT	PK
employee_id	INT	FK(Employee)
shift_id	INT	FK(Shift)
attendance_date	DATE	NOT NULL
check_in	TIME	NULL
lunch_in	TIME	NULL
lunch_out	TIME	NULL
check_out	TIME	NULL
worked_hours	DECIMAL(5,2)	DEFAULT 0
overtime_hours	DECIMAL(5,2)	DEFAULT 0
late_minutes	INT	DEFAULT 0
early_exit_minutes	INT	DEFAULT 0

Column Name	Data Type	Constraint
attendance_status	ENUM('Present','Absent','Half Day','Leave','Holiday','Weekend','WFH')	NOT NULL
remarks	VARCHAR(255)	NULL

Table ATT004 : attendance_regularization

Column Name	Data Type	Constraint
request_id	INT	PK
employee_id	INT	FK
attendance_date	DATE	NOT NULL
corrected_check_in	TIME	NULL
corrected_check_out	TIME	NULL
reason	TEXT	NOT NULL
manager_id	INT	FK(Employee)
hr_id	INT	FK(Employee)
status	ENUM('Pending','Approved','Rejected')	Default Pending

Table ATT005 : overtime

Column Name	Data Type	Constraint
overtime_id	INT	PK
employee_id	INT	FK
attendance_id	INT	FK

Column Name	Data Type	Constraint
overtime_hours	DECIMAL(5,2)	NOT NULL
approved_by	INT	FK(Employee)
status	ENUM('Pending','Approved','Rejected')	Default Pending

Table ATT006 : timesheet

Column Name	Data Type	Constraint
timesheet_id	INT	PK
employee_id	INT	FK(Employee)
project_id	INT	FK(Project)
task_id	INT	FK(Task)
work_date	DATE	NOT NULL
worked_hours	DECIMAL(5,2)	NOT NULL
description	TEXT	NULL
status	ENUM('Draft','Submitted','Approved','Rejected')	Default Draft
approved_by	INT	FK(Employee)

Entity Relationship

Shift Master (1)



Attendance Policy (1)

|
└──────────< Shift Master

Employee (1)

|──────────< Employee Attendance
|──────────< Attendance Regularization
|──────────< Overtime
└──────────< Timesheet

Project (1)

|
└──────────< Timesheet

Task (1)

|
└──────────< Timesheet

Dashboard Widgets

- Today's Attendance
- Present Employees
- Absent Employees
- Late Arrivals
- Employees on Leave
- Overtime Hours
- Pending Regularization Requests
- Pending Timesheet Approvals

- Monthly Attendance Percentage
 - Attendance Trend Chart
-

Deliverables

- Shift Master
 - Attendance Policy
 - Employee Attendance
 - Attendance Regularization
 - Overtime Management
 - Timesheet Management
 - Attendance Reports
 - Attendance Dashboard
 - Payroll Integration
-

7. Leave Management Module

7.1 Module Overview

The Leave Management Module automates the complete employee leave lifecycle, including leave type configuration, leave allocation, leave policy management, leave application, approval workflow, leave balance tracking, leave encashment, carry forward, and leave reports.

The module integrates with the Attendance and Payroll modules to ensure accurate leave calculations, salary processing, and employee attendance records.

7.2 Module Objectives

The Leave Management Module aims to:

- Maintain different leave types.
 - Allocate leaves department-wise.
 - Configure company leave policies.
 - Allow employees to apply for leave online.
 - Enable Manager and HR approval workflow.
 - Track leave balances.
 - Support leave carry forward and encashment.
 - Integrate leave records with attendance and payroll.
 - Generate leave reports.
-

7.3 User Roles

User Role	Permission
Administrator	Full Access
HR Manager	Full Access

User Role	Permission
HR Executive	Manage Leave Policies
Team Manager	Approve/Reject Leave
Employee	Apply & View Leave

7.4 Module Components

The Leave Management Module consists of:

1. Leave Type Master
2. Leave Allocation
3. Leave Policy / Leave Settings
4. Apply Leave
5. Leave Approval Workflow
6. Leave Balance
7. Leave Cancellation
8. Leave Encashment
9. Leave Reports

7.5 Leave Workflow

Create Leave Type

|

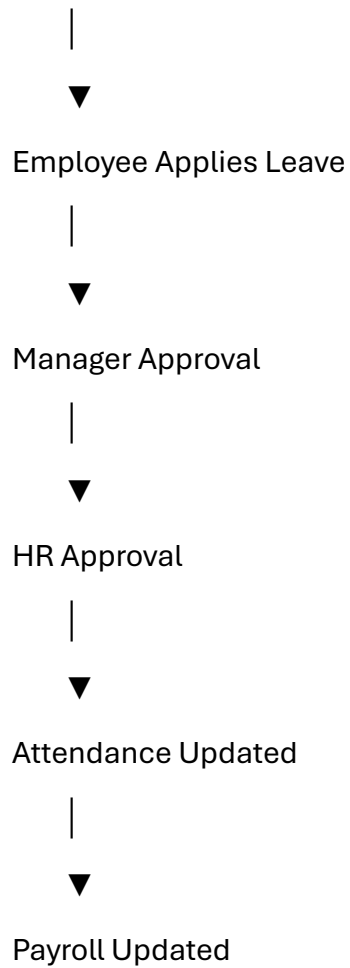


Allocate Leave Department-wise

|



Configure Leave Policy



Functional Requirements

FR-LV-001

Leave Type Master

Description

The Leave Type Master is used to define various leave categories available within the organization.

Functions

- Add Leave Type

- Update Leave Type
- Delete Leave Type
- Activate Leave Type
- Search Leave Type
- View Leave Type List

Leave Types

- Casual Leave (CL)
- Sick Leave (SL)
- Privilege Leave (PL)
- Earned Leave (EL)
- Maternity Leave
- Paternity Leave
- Comp Off
- Work From Home
- Leave Without Pay (LWP)

Leave Type Details

Field	Type
Leave Type ID	Auto
Leave Code	Text
Leave Name	Text
Description	TextArea
Paid Leave	Yes/No

Field	Type
Carry Forward Allowed	Yes/No
Encashment Allowed	Yes/No
Maximum Days	Number
Status	Active/Inactive

Business Rules

- Leave Code must be unique.
- Leave Name must be unique.
- Maximum Leave Days cannot be negative.

FR-LV-002

Leave Allocation

Description

Allocate leave balances to employees or departments annually.

Functions

- Allocate Department-wise Leave
- Allocate Employee-wise Leave
- Update Allocation
- Reset Yearly Leave
- View Allocation

Leave Allocation Details

Field	Type
Allocation ID	Auto
Department	Dropdown
Leave Type	Dropdown
Total Leave	Number
Effective From	Date
Effective To	Date
Status	Active

Example Allocation

Department	CL	SL	PL
IT	12	10	18
HR	10	10	15
Finance	10	8	15

Business Rules

- Allocation is effective for a defined period.
- Existing allocation must expire before a new allocation is created.

FR-LV-003

Leave Policy / Leave Settings

Description

Configure organizational leave rules.

Leave Settings

- Weekly Off
- Public Holidays
- Sandwich Leave
- Half Day Leave
- Minimum Leave Duration
- Maximum Continuous Leave
- Carry Forward Policy
- Leave Encashment
- Leave Expiry
- Probation Leave Rule
- Notice Period Leave Rule

Example Policy

Setting	Value
Weekly Off	Saturday, Sunday
Carry Forward	Yes
Maximum Carry Forward	10 Days
Leave Encashment	Yes
Sandwich Rule	No
Half Day Allowed	Yes
Minimum Leave	0.5 Day

FR-LV-004

Apply Leave

Description

Employees can apply for leave through the HRMS portal.

Functions

- Apply Leave
 - Edit Leave (Before Approval)
 - Cancel Leave
 - View Leave Status
 - View Leave History
-

Leave Application Details

Field	Type
Leave Application ID	Auto
Employee	Auto
Leave Type	Dropdown
From Date	Date
To Date	Date
Number of Days	Auto Calculate
Half Day	Yes/No
Reason	TextArea
Attachment	Upload (Medical Certificate etc.)

Field	Type
Status	Pending/Approved/Rejected

Business Rules

- Leave balance must be available.
- From Date cannot be greater than To Date.
- Past leave applications require HR approval.
- Medical certificate is mandatory for Sick Leave above company policy limit.

FR-LV-005

Leave Approval Workflow

Description

Managers and HR approve employee leave requests.

Workflow

Employee

↓

Manager

↓

HR

↓

Attendance Updated

↓

Payroll Updated

Approval Status

- Pending
 - Manager Approved
 - HR Approved
 - Rejected
 - Cancelled
-

Business Rules

- Manager approval is mandatory.
 - HR can override Manager approval only with remarks.
 - Approved leave updates Attendance automatically.
-

FR-LV-006

Leave Balance

Description

Displays employee leave availability.

Leave Balance Details

Field	Type
Employee	Auto
Leave Type	Auto
Allocated	Number
Availed	Number
Remaining	Auto
Carry Forward	Number
Encashed	Number

FR-LV-007

Leave Encashment

Description

Employees can encash eligible leave balances according to company policy.

Encashment Details

Field	Type
Request ID	Auto
Employee	Dropdown
Leave Type	Dropdown
Available Leave	Auto
Requested Days	Number
Amount	Auto Calculate

Field	Type
Status	Pending/Approved/Rejected

Business Rules

- Only eligible leave types can be encashed.
- Encashed leave reduces leave balance.
- Payroll processes the encashment amount.

FR-LV-008

Leave Reports

The Leave Module provides the following reports:

- Leave Type Report
- Leave Allocation Report
- Leave Balance Report
- Employee Leave History
- Monthly Leave Report
- Department-wise Leave Report
- Leave Encashment Report
- Pending Leave Approval Report
- Rejected Leave Report
- Leave Summary Report

Database Design

Table LV001 : leave_type

Column Name	Data Type	Constraint
leave_type_id	INT	PK
leave_code	VARCHAR(20)	UNIQUE
leave_name	VARCHAR(100)	NOT NULL
description	VARCHAR(255)	NULL
is_paid	BOOLEAN	Default TRUE
carry_forward	BOOLEAN	Default FALSE
encashment_allowed	BOOLEAN	Default FALSE
maximum_days	INT	NOT NULL
status	ENUM('Active','Inactive')	Default Active

Table LV002 : leave_allocation

Column Name	Data Type	Constraint
allocation_id	INT	PK
department_id	INT	FK(Department)
leave_type_id	INT	FK(Leave Type)
total_leave	DECIMAL(5,1)	NOT NULL
effective_from	DATE	NOT NULL
effective_to	DATE	NOT NULL
status	ENUM('Active','Inactive')	Default Active

Table LV003 : leave_policy

Column Name	Data Type	Constraint
policy_id	INT	PK
weekly_off	VARCHAR(100)	NOT NULL
carry_forward_limit	INT	DEFAULT 0
encashment_allowed	BOOLEAN	DEFAULT FALSE
sandwich_rule	BOOLEAN	DEFAULT FALSE
half_day_allowed	BOOLEAN	DEFAULT TRUE
minimum_leave	DECIMAL(3,1)	DEFAULT 0.5
maximum_continuous_leave	INT	DEFAULT 30
probation_leave_allowed	BOOLEAN	DEFAULT FALSE

Table LV004 : leave_application

Column Name	Data Type	Constraint
leave_application_id	INT	PK
employee_id	INT	FK(Employee)
leave_type_id	INT	FK
from_date	DATE	NOT NULL
to_date	DATE	NOT NULL
total_days	DECIMAL(4,1)	NOT NULL
half_day	BOOLEAN	DEFAULT FALSE
reason	TEXT	NOT NULL
attachment	VARCHAR(255)	NULL

Column Name	Data Type	Constraint
manager_id	INT	FK(Employee)
hr_id	INT	FK(Employee)
status	ENUM('Pending','Manager Approved','HR Approved','Rejected','Cancelled')	Default Pending
applied_date	DATETIME	CURRENT_TIMESTAMP

Table LV005 : leave_balance

Column Name	Data Type	Constraint
balance_id	INT	PK
employee_id	INT	FK(Employee)
leave_type_id	INT	FK
allocated_leave	DECIMAL(5,1)	DEFAULT 0
availed_leave	DECIMAL(5,1)	DEFAULT 0
carry_forward_leave	DECIMAL(5,1)	DEFAULT 0
encashed_leave	DECIMAL(5,1)	DEFAULT 0
remaining_leave	DECIMAL(5,1)	DEFAULT 0

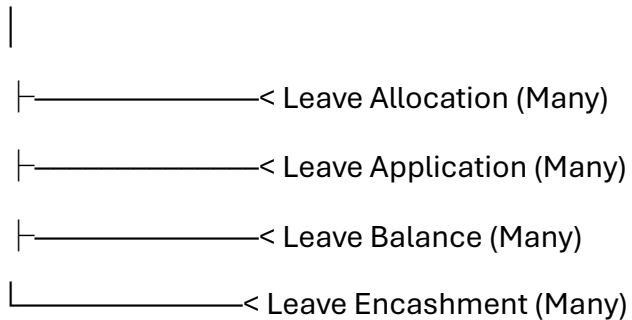
Table LV006 : leave_encashment

Column Name	Data Type	Constraint
encashment_id	INT	PK
employee_id	INT	FK(Employee)

Column Name	Data Type	Constraint
leave_type_id	INT	FK
available_leave	DECIMAL(5,1)	NOT NULL
requested_leave	DECIMAL(5,1)	NOT NULL
amount	DECIMAL(10,2)	NOT NULL
approved_by	INT	FK(Employee)
status	ENUM('Pending','Approved','Rejected')	Default Pending

Entity Relationship

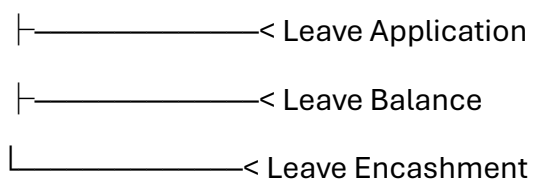
Leave Type (1)



Department (1)



Employee (1)



Dashboard Widgets

- Employees on Leave Today
- Pending Leave Approvals
- Leave Balance Summary
- Monthly Leave Statistics
- Department-wise Leave Count
- Upcoming Holidays
- Leave Calendar
- Leave Trend Chart

Deliverables

- Leave Type Master
 - Leave Allocation
 - Leave Policy Management
 - Apply Leave
 - Leave Approval Workflow
 - Leave Balance
 - Leave Encashment
 - Leave Reports
 - Leave Dashboard
 - Attendance & Payroll Integration
-

8. Payroll Management Module

8.1 Module Overview

The Payroll Management Module is responsible for calculating employee salaries based on attendance, leave, earnings, deductions, overtime, incentives, and statutory deductions. It automates monthly payroll processing, salary slip generation, payroll transactions, and salary history.

The Payroll module integrates with the Employee, Attendance, Leave, and Finance modules to ensure accurate salary calculations and compliance with company policies.

8.2 Module Objectives

The Payroll Management Module aims to:

- Maintain salary structures.
 - Configure earnings and deductions.
 - Process monthly payroll automatically.
 - Generate employee salary slips.
 - Maintain payroll transaction history.
 - Integrate attendance and leave data.
 - Calculate overtime and leave deductions.
 - Generate payroll reports.
 - Maintain complete salary history.
-

8.3 User Roles

User Role	Permission
Administrator	Full Access
HR Manager	Full Access

User Role	Permission
HR Executive	Process Payroll
Finance Manager	Verify Payroll
Employee	View Salary Slip & Salary History

8.4 Module Components

The Payroll Module consists of:

1. Earning Master
2. Deduction Master
3. Salary Structure
4. Employee Salary
5. Payroll Processing
6. Salary Slip Generation
7. Payroll Transaction History
8. Payroll Reports

8.5 Payroll Workflow

Create Earnings

|



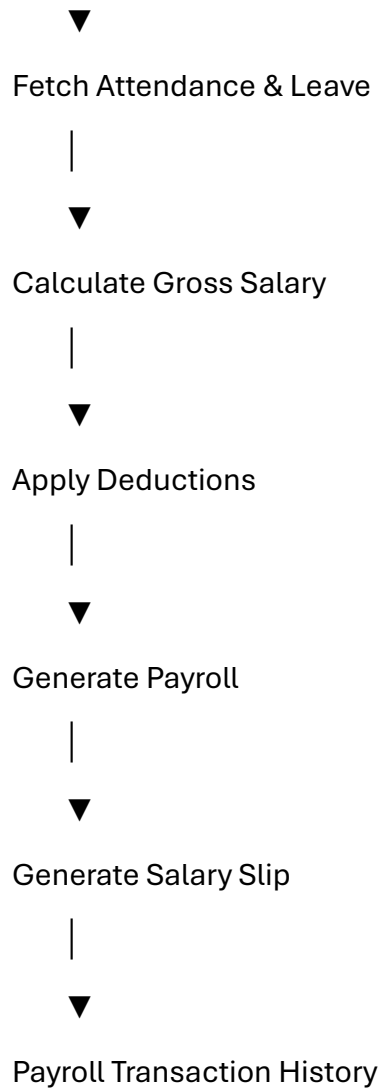
Create Deductions

|



Assign Salary Structure

|



Functional Requirements

FR-PAY-001

Earning Master

Description

The Earning Master stores all salary components that increase the employee's gross salary.

Functions

- Add Earning
- Update Earning
- Delete Earning
- Activate / Deactivate
- Search Earnings

Sample Earnings

- Basic Salary
- House Rent Allowance (HRA)
- Dearness Allowance (DA)
- Medical Allowance
- Travel Allowance
- Conveyance Allowance
- Special Allowance
- Performance Bonus
- Incentive
- Overtime Amount

Earning Details

Field	Type
Earning ID	Auto
Earning Name	Text
Earning Code	Text
Calculation Type	Fixed / Percentage

Field	Type
Amount	Decimal
Taxable	Yes / No
Status	Active

Business Rules

- Earning Code must be unique.
 - Amount cannot be negative.
 - Percentage should be between 0 and 100.
-

FR-PAY-002

Deduction Master

Description

Stores salary deduction components.

Sample Deductions

- Provident Fund (PF)
 - Professional Tax (PT)
 - Employee State Insurance (ESI)
 - Income Tax (TDS)
 - Loan Recovery
 - Advance Salary Recovery
 - Leave Without Pay (LWP)
 - Other Deduction
-

Deduction Details

Field	Type
Deduction ID	Auto
Deduction Name	Text
Deduction Code	Text
Calculation Type	Fixed / Percentage
Amount	Decimal
Status	Active

Business Rules

- Deduction Code must be unique.
- Deduction Amount cannot be negative.

FR-PAY-003

Salary Structure

Description

Defines salary components assigned to employees.

Functions

- Create Salary Structure
- Edit Salary Structure
- View Salary Structure

Salary Structure Details

Field	Type
Salary Structure ID	Auto
Employee	Dropdown
Effective Date	Date
Basic Salary	Decimal
Gross Salary	Decimal
Net Salary	Auto Calculate
Status	Active / Inactive

Salary Components

Earnings

- Basic Salary
- HRA
- DA
- Medical
- Conveyance
- Bonus
- Incentive
- Overtime

Deductions

- PF
- PT
- ESI
- TDS

- LWP
- Loan
- Advance
- Other Deduction

FR-PAY-004

Employee Salary

Description

Stores monthly salary information for each employee.

Salary Details

Field	Type
Employee	Dropdown
Payroll Month	Month
Payroll Year	Year
Gross Salary	Decimal
Total Earnings	Decimal
Total Deductions	Decimal
Net Salary	Decimal
Payment Status	Pending / Paid

FR-PAY-005

Payroll Processing

Description

Processes employee salaries every month.

Payroll Process

1. Select Payroll Month
 2. Fetch Attendance
 3. Fetch Leave
 4. Fetch Salary Structure
 5. Calculate Earnings
 6. Calculate Deductions
 7. Calculate Net Salary
 8. Generate Salary Slip
 9. Lock Payroll
-

Business Rules

- Payroll can be processed only once per month.
 - Locked payroll cannot be modified without authorization.
 - Attendance and approved leave are considered during payroll.
-

FR-PAY-006

Salary Slip Generation

Description

Generates monthly salary slips for employees.

Salary Slip Includes

Employee Information

- Employee ID

- Employee Name
 - Department
 - Designation
 - Bank Account
 - UAN Number
 - PAN Number
-

Attendance Summary

- Working Days
 - Present Days
 - Paid Leave
 - LWP
 - Overtime Hours
-

Earnings

- Basic Salary
 - HRA
 - DA
 - Medical
 - Special Allowance
 - Incentive
 - Overtime
-

Deductions

- PF
- PT

- ESI
- TDS
- Loan
- LWP
- Other Deduction

Salary Summary

- Gross Salary
- Total Deduction
- Net Salary

FR-PAY-007

Payroll Transaction History

Description

Stores complete payroll history for every employee.

Transaction Details

Field	Type
Transaction ID	Auto
Employee	Dropdown
Payroll Month	Month
Salary Paid Date	Date
Gross Salary	Decimal
Deduction	Decimal

Field	Type
Net Salary	Decimal
Payment Mode	Bank Transfer / Cash / Cheque
Transaction Reference	Text
Payment Status	Paid / Pending / Failed

Payroll Reports

The Payroll Module provides the following reports:

- Employee Salary Report
- Monthly Payroll Report
- Salary Register
- PF Report
- PT Report
- TDS Report
- Overtime Report
- Deduction Report
- Payroll Transaction Report
- Salary Slip Report

Database Design

Table PAY001 : earning_master

Column Name	Data Type	Constraint
earning_id	INT	PK

Column Name	Data Type	Constraint
earning_name	VARCHAR(100)	NOT NULL
earning_code	VARCHAR(20)	UNIQUE
calculation_type	ENUM('Fixed','Percentage')	NOT NULL
amount	DECIMAL(10,2)	NOT NULL
taxable	BOOLEAN	DEFAULT TRUE
status	ENUM('Active','Inactive')	DEFAULT 'Active'

Table PAY002 : deduction_master

Column Name	Data Type	Constraint
deduction_id	INT	PK
deduction_name	VARCHAR(100)	NOT NULL
deduction_code	VARCHAR(20)	UNIQUE
calculation_type	ENUM('Fixed','Percentage')	NOT NULL
amount	DECIMAL(10,2)	NOT NULL
status	ENUM('Active','Inactive')	DEFAULT 'Active'

Table PAY003 : salary_structure

Column Name	Data Type	Constraint
salary_structure_id	INT	PK
employee_id	INT	FK(Employee)

Column Name	Data Type	Constraint
effective_date	DATE	NOT NULL
basic_salary	DECIMAL(10,2)	NOT NULL
gross_salary	DECIMAL(10,2)	NOT NULL
net_salary	DECIMAL(10,2)	NOT NULL
status	ENUM('Active','Inactive')	DEFAULT 'Active'

Table PAY004 : employee_salary

Column Name	Data Type	Constraint
employee_salary_id	INT	PK
employee_id	INT	FK(Employee)
payroll_month	INT	NOT NULL
payroll_year	INT	NOT NULL
working_days	INT	NOT NULL
present_days	INT	NOT NULL
leave_days	DECIMAL(5,1)	DEFAULT 0
lwp_days	DECIMAL(5,1)	DEFAULT 0
overtime_hours	DECIMAL(5,2)	DEFAULT 0
total_earnings	DECIMAL(12,2)	NOT NULL
total_deductions	DECIMAL(12,2)	NOT NULL
gross_salary	DECIMAL(12,2)	NOT NULL

Column Name	Data Type	Constraint
net_salary	DECIMAL(12,2)	NOT NULL
payroll_status	ENUM('Draft','Processed','Paid')	DEFAULT 'Draft'

Table PAY005 : salary_slip

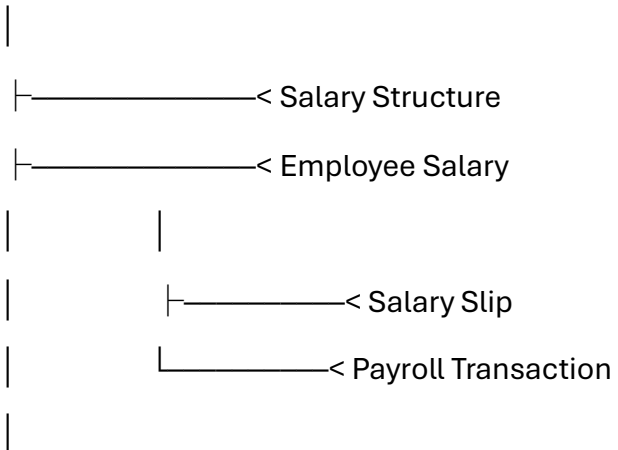
Column Name	Data Type	Constraint
salary_slip_id	INT	PK
employee_salary_id	INT	FK(Employee Salary)
salary_slip_number	VARCHAR(50)	UNIQUE
generated_date	DATETIME	CURRENT_TIMESTAMP
pdf_path	VARCHAR(255)	NULL
email_sent	BOOLEAN	DEFAULT FALSE

Table PAY006 : payroll_transaction

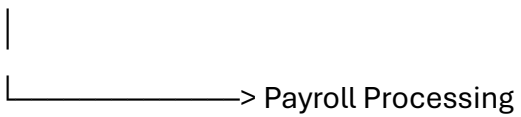
Column Name	Data Type	Constraint
transaction_id	INT	PK
employee_salary_id	INT	FK(Employee Salary)
payment_date	DATE	NOT NULL
payment_mode	ENUM('Bank Transfer','Cash','Cheque','UPI')	NOT NULL
transaction_reference	VARCHAR(100)	NULL
paid_amount	DECIMAL(12,2)	NOT NULL
payment_status	ENUM('Pending','Paid','Failed')	DEFAULT 'Pending'

Entity Relationship

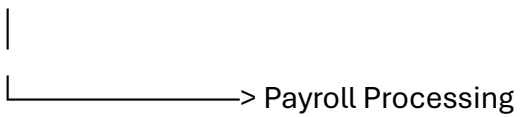
Employee (1)



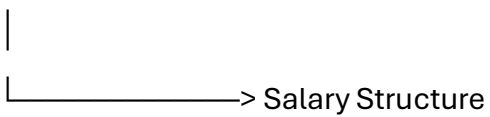
Attendance (Many)



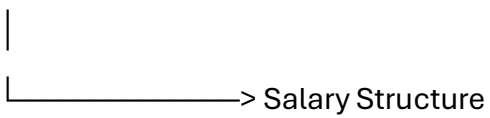
Leave (Many)



Earning Master (1)



Deduction Master (1)



Payroll Dashboard

The Payroll Dashboard displays:

- Total Employees Paid
 - Pending Payroll
 - Total Gross Salary
 - Total Net Salary
 - Total Earnings
 - Total Deductions
 - Overtime Amount
 - LWP Amount
 - Monthly Payroll Summary
 - Payroll Trend Chart
-

Business Rules

- Payroll is processed once per payroll cycle.
 - Salary cannot be modified after payroll is locked.
 - Only approved attendance and leave records are considered.
 - Salary slips are generated only after payroll processing.
 - Every salary payment creates a payroll transaction.
 - Payroll data is retained permanently for audit purposes.
-

Deliverables

- Earning Master
- Deduction Master
- Salary Structure
- Employee Salary Management

- Payroll Processing
 - Salary Slip Generation
 - Payroll Transaction History
 - Payroll Reports
 - Payroll Dashboard
 - Attendance & Leave Integration
-

9. Project Management Module

9.1 Module Overview

The Project Management Module enables organizations to create projects, assign employees, manage tasks, monitor project progress, track milestones, and visualize work using a Kanban Task Board. This module integrates with the Timesheet Module to record project-wise working hours and provides real-time visibility into project status.

The module helps Project Managers efficiently plan, execute, monitor, and complete projects while ensuring accountability and collaboration among team members.

9.2 Module Objectives

The Project Management Module aims to:

- Create and manage projects.
- Allocate employees to projects.
- Create and assign tasks.
- Track project milestones.
- Monitor project progress.
- Maintain task comments and attachments.
- Integrate with employee timesheets.

- Provide project dashboards and reports.

9.3 User Roles

User Role	Permission
Administrator	Full Access
Project Manager	Create & Manage Projects
Team Lead	Assign Tasks
HR Manager	View Projects
Employee	View Assigned Projects & Update Tasks

9.4 Module Components

The Project Management Module consists of:

1. Project Master
2. Project Team Allocation
3. Task Management
4. Task Assignment
5. Task Board (Kanban)
6. Project Milestones
7. Task Comments
8. Task Attachments
9. Project Reports

9.5 Project Workflow

Create Project

|

▼
Allocate Team Members

|
▼
Create Milestones

|
▼
Create Tasks

|
▼
Assign Tasks

|
▼
Update Task Status

|
▼
Submit Timesheet

|
▼
Project Completion

Functional Requirements

FR-PRJ-001

Project Master

Description

The Project Master stores all project-related information including project details, client information, project duration, budget, and project status.

Functions

- Add Project
 - Edit Project
 - Delete Project
 - Search Project
 - View Project Details
 - Close Project
-

Project Details

Field	Type
Project ID	Auto
Project Code	Text
Project Name	Text
Client Name	Text
Project Manager	Dropdown
Department	Dropdown
Start Date	Date
End Date	Date
Budget	Decimal
Priority	Low / Medium / High
Project Status	Planning / Active / On Hold / Completed / Cancelled

Field	Type
Description	TextArea

Business Rules

- Project Code must be unique.
 - End Date must be greater than Start Date.
 - Closed projects cannot be edited.
-

FR-PRJ-002

Project Team Allocation

Description

Assign employees to projects.

Functions

- Assign Employee
 - Remove Employee
 - Change Project Role
 - View Team Members
-

Allocation Details

Field	Type
Allocation ID	Auto
Project	Dropdown
Employee	Dropdown

Field	Type
Role	Developer / Tester / Designer / Team Lead
Allocation Date	Date
Release Date	Date
Status	Active / Released

Business Rules

- One employee can work on multiple projects.
- Duplicate allocation is not allowed.
- Released employees cannot update project tasks.

FR-PRJ-003

Task Management

Description

Manage project tasks.

Functions

- Add Task
 - Update Task
 - Delete Task
 - Assign Task
 - Change Status
 - Add Comments
 - Upload Attachment
-

Task Details

Field	Type
Task ID	Auto
Project	Dropdown
Task Name	Text
Task Description	TextArea
Milestone	Dropdown
Assigned To	Employee
Priority	Low / Medium / High
Start Date	Date
Due Date	Date
Estimated Hours	Decimal
Status	To Do / In Progress / Review / Completed

Business Rules

- Due Date must be after Start Date.
- Estimated Hours must be greater than zero.
- Completed tasks cannot be modified without manager approval.

FR-PRJ-004

Task Assignment

Description

Assign project tasks to employees.

Workflow

Manager



Assign Task



Employee



Accept Task



Start Work



Complete Task

Assignment Details

Field	Type
Assignment ID	Auto
Task	Dropdown
Employee	Dropdown
Assigned By	Manager
Assigned Date	Date
Status	Assigned / Accepted / Rejected / Completed

Business Rules

- One task can have only one primary owner.
 - Task reassignment requires manager approval.
-

FR-PRJ-005

Kanban Task Board

Description

The Kanban Board provides a visual representation of task progress.

Columns

- Backlog
 - To Do
 - In Progress
 - Code Review
 - Testing
 - Completed
-

Features

- Drag & Drop Tasks
 - Task Progress
 - Task Priority Color
 - Due Date Indicator
 - Assignee Avatar
 - Progress Percentage
-

FR-PRJ-006

Project Milestones

Description

Milestones divide the project into measurable phases.

Milestone Details

Field	Type
Milestone ID	Auto
Project	Dropdown
Milestone Name	Text
Start Date	Date
End Date	Date
Status	Pending / In Progress / Completed

Business Rules

- Milestone dates must fall within project duration.

- Milestone completion depends on associated tasks.

FR-PRJ-007

Task Comments

Description

Allows collaboration among project team members.

Features

- Add Comment
 - Edit Comment
 - Delete Comment
 - Mention Employee
 - Reply to Comment
-

Comment Details

Field	Type
Comment ID	Auto
Task	Dropdown
Employee	Dropdown
Comment	TextArea
Comment Date	DateTime

FR-PRJ-008

Task Attachments

Description

Upload documents related to project tasks.

Supported Files

- PDF
 - DOCX
 - XLSX
 - JPG
 - PNG
 - ZIP
-

Attachment Details

Field	Type
Attachment ID	Auto
Task	Dropdown
File Name	Text
File Path	Text
Uploaded By	Employee
Upload Date	DateTime

Reports

The Project Module provides the following reports:

- Project Master Report
- Active Projects Report

- Completed Projects Report
- Employee-wise Project Report
- Team Allocation Report
- Task Status Report
- Overdue Task Report
- Milestone Progress Report
- Timesheet vs Estimated Hours Report
- Project Budget Report

Database Design

Table PRJ001 : project

Column Name	Data Type	Constraint
project_id	INT	PK
project_code	VARCHAR(20)	UNIQUE
project_name	VARCHAR(150)	NOT NULL
client_name	VARCHAR(150)	NOT NULL
manager_id	INT	FK(Employee)
department_id	INT	FK(Department)
start_date	DATE	NOT NULL
end_date	DATE	NOT NULL
budget	DECIMAL(15,2)	NULL
priority	ENUM('Low','Medium','High')	DEFAULT 'Medium'

Column Name	Data Type	Constraint
project_status	ENUM('Planning','Active','On Hold','Completed','Cancelled')	DEFAULT 'Planning'
description	TEXT	NULL

Table PRJ002 : project_team

Column Name	Data Type	Constraint
team_id	INT	PK
project_id	INT	FK(Project)
employee_id	INT	FK(Employee)
project_role	VARCHAR(100)	NOT NULL
allocation_date	DATE	NOT NULL
release_date	DATE	NULL
status	ENUM('Active','Released')	DEFAULT 'Active'

Table PRJ003 : project_milestone

Column Name	Data Type	Constraint
milestone_id	INT	PK
project_id	INT	FK(Project)
milestone_name	VARCHAR(150)	NOT NULL
start_date	DATE	NOT NULL

Column Name	Data Type	Constraint
end_date	DATE	NOT NULL
status	ENUM('Pending','In Progress','Completed')	DEFAULT 'Pending'

Table PRJ004 : task

Column Name	Data Type	Constraint
task_id	INT	PK
project_id	INT	FK(Project)
milestone_id	INT	FK(Project Milestone)
task_name	VARCHAR(150)	NOT NULL
task_description	TEXT	NULL
assigned_to	INT	FK(Employee)
priority	ENUM('Low','Medium','High')	DEFAULT 'Medium'
estimated_hours	DECIMAL(5,2)	NOT NULL
start_date	DATE	NOT NULL
due_date	DATE	NOT NULL
task_status	ENUM('Backlog','To Do','In Progress','Code Review','Testing','Completed')	DEFAULT 'Backlog'

Table PRJ005 : task_comment

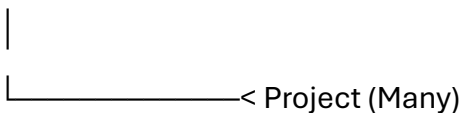
Column Name	Data Type	Constraint
comment_id	INT	PK
task_id	INT	FK(Task)
employee_id	INT	FK(Employee)
comment	TEXT	NOT NULL
commented_on	DATETIME	CURRENT_TIMESTAMP

Table PRJ006 : task_attachment

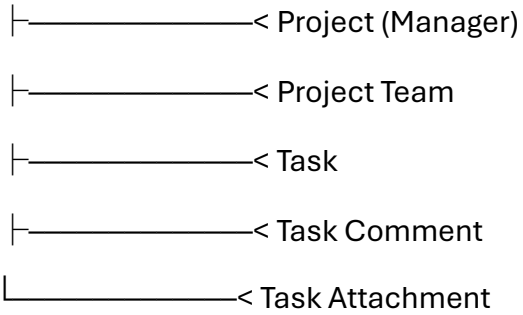
Column Name	Data Type	Constraint
attachment_id	INT	PK
task_id	INT	FK(Task)
file_name	VARCHAR(255)	NOT NULL
file_path	VARCHAR(255)	NOT NULL
uploaded_by	INT	FK(Employee)
uploaded_on	DATETIME	CURRENT_TIMESTAMP

Entity Relationship

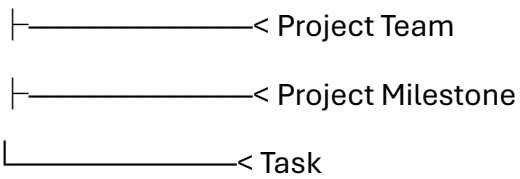
Department (1)



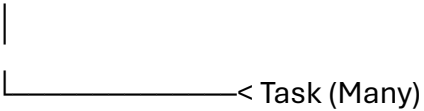
Employee (1)



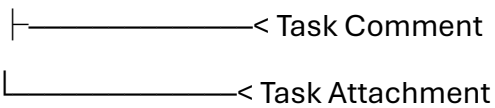
Project (1)



Project Milestone (1)



Task (1)



Dashboard Widgets

- Total Projects
- Active Projects
- Completed Projects
- Delayed Projects
- Upcoming Milestones
- Pending Tasks

- Overdue Tasks
 - Team Workload
 - Project Progress (%)
 - Timesheet Summary
-

Business Rules

- Every project must have one Project Manager.
 - Employees must be allocated to a project before task assignment.
 - Project status is updated automatically based on milestone completion (optional configuration).
 - Completed projects are archived but remain available for reporting.
 - Timesheet entries must reference valid projects and tasks.
 - File attachments are restricted to configured file types and maximum upload size.
-

Deliverables

- Project Master
 - Project Team Allocation
 - Project Milestone Management
 - Task Management
 - Task Assignment
 - Kanban Task Board
 - Task Comments
 - Task Attachments
 - Project Reports
 - Project Dashboard
-

10. Document Management Module

10.1 Module Overview

The Document Management Module is designed to manage all company and employee-related documents in a centralized digital repository. It allows administrators to maintain document categories, upload organization documents, verify employee documents, track document expiry, and provide secure access based on user roles.

This module eliminates manual file handling, improves document security, and ensures easy retrieval of HR-related documents.

10.2 Module Objectives

The Document Management Module aims to:

- Maintain master document categories.
- Store company HR documents.
- Store employee personal and educational documents.
- Support secure document upload and download.
- Verify employee documents.
- Track document expiry dates.
- Maintain document version history.
- Generate document-related reports.

10.3 User Roles

User Role	Permission
Administrator	Full Access
HR Manager	Full Access
HR Executive	Upload & Verify Documents
Employee	Upload/View Own Documents
Manager	View Team Employee Documents (Restricted)

10.4 Module Components

The Document Management Module consists of:

1. Master Document Type
2. Admin Documents
3. Employee Documents
4. Upload Document
5. Document Verification
6. Document Expiry Tracking
7. Document Reports

10.5 Document Workflow

Create Document Type



Upload Document



Store File Metadata



Document Verification



Approve / Reject



Employee Notification





Document Available for Download

Functional Requirements

FR-DOC-001

Master Document Type

Description

Defines different categories of documents used in the HRMS application.

Functions

- Add Document Type
- Edit Document Type
- Delete Document Type
- Activate / Deactivate
- Search Document Type

Document Type Details

Field	Type
Document Type ID	Auto
Document Type Name	Text
Applicable For	Admin / Employee / Both
Description	TextArea
Is Mandatory	Yes / No
Expiry Required	Yes / No
Status	Active / Inactive

Sample Document Types

Admin Documents

- Offer Letter
- Appointment Letter
- Experience Letter
- Salary Slip Template
- HR Policy
- Company Circular
- Employee Handbook
- Increment Letter

Employee Documents

- Aadhaar Card
- PAN Card
- Passport
- Driving License
- Resume
- Graduation Certificate
- Post Graduation Certificate
- Experience Certificate
- Bank Passbook
- Medical Certificate

Business Rules

- Document Type Name must be unique.
- Mandatory documents are required during employee onboarding.
- Expiry tracking is enabled only when configured.

FR-DOC-002

Admin Documents

Description

Stores organization-level HR and company documents.

Functions

- Upload Document
- Update Document
- Delete Document
- Download Document
- Publish / Unpublish

Document Details

Field	Type
Document ID	Auto
Document Type	Dropdown
Document Name	Text
Description	TextArea
File Upload	File
Version Number	Number
Effective Date	Date
Expiry Date	Date
Access Level	HR / Manager / Employee / All
Status	Draft / Published / Archived

Business Rules

- Published documents are visible according to Access Level.
- Archived documents remain available for audit purposes.
- Version history is maintained for updated documents.

FR-DOC-003

Employee Documents

Description

Stores employee-uploaded personal, educational, identity, and employment documents.

Functions

- Upload Document
- Replace Document
- View Document
- Download Document
- Delete Document (Before Verification)

Employee Document Details

Field	Type
Employee Document ID	Auto
Employee	Auto
Document Type	Dropdown
Document Number	Text
File Upload	File
Issue Date	Date
Expiry Date	Date
Remarks	TextArea
Verification Status	Pending / Verified / Rejected

Business Rules

- Employees can view only their own documents.
- Aadhaar and PAN documents must be unique.

- Verified documents cannot be deleted by employees.

FR-DOC-004

Upload Document

Description

Provides a common file upload mechanism for both Admin and Employee documents.

Supported File Types

- PDF
- DOC
- DOCX
- JPG
- JPEG
- PNG
- XLS
- XLSX

File Validation

Validation	Value
Maximum File Size	10 MB
Allowed Extensions	Configurable
Virus Scan	Optional
Duplicate File Check	Yes

Upload Rules

- File name is auto-generated to avoid conflicts.
- Original file name is stored separately.
- Upload date and uploaded user are recorded.

FR-DOC-005

Document Verification

Description

HR verifies employee-uploaded documents.

Verification Workflow

Employee Uploads Document

↓

HR Review

↓

Approve / Reject

↓

Employee Notification

Verification Status

- Pending
- Verified
- Rejected

Business Rules

- Only HR/Admin can verify documents.
- Rejected documents require remarks.
- Employees are notified after verification.

FR-DOC-006

Document Expiry Tracking

Description

Tracks documents with expiry dates such as Passport, Driving License, Medical Certificate, etc.

Features

- Expiry Date Monitoring
- Upcoming Expiry Alerts
- Expired Document Report
- Email Notifications

Alert Configuration

Alert Period	Notification
30 Days Before Expiry	Email + Dashboard
15 Days Before Expiry	Email + Dashboard
7 Days Before Expiry	Email + Dashboard
On Expiry Date	Dashboard Alert

Reports

The Document Module provides:

- Document Type Report
- Employee Document Report
- Verified Document Report
- Pending Verification Report
- Rejected Document Report
- Expired Document Report
- Upcoming Expiry Report
- Admin Document Report

- Mandatory Document Missing Report

Database Design

Table DOC001 : document_type

Column Name	Data Type	Constraint
document_type_id	INT	PK
document_type_name	VARCHAR(100)	UNIQUE
applicable_for	ENUM('Admin','Employee','Both')	NOT NULL
description	VARCHAR(255)	NULL
is_mandatory	BOOLEAN	DEFAULT FALSE
expiry_required	BOOLEAN	DEFAULT FALSE
status	ENUM('Active','Inactive')	DEFAULT 'Active'

Table DOC002 : admin_document

Column Name	Data Type	Constraint
admin_document_id	INT	PK
document_type_id	INT	FK(Document Type)
document_name	VARCHAR(150)	NOT NULL
description	TEXT	NULL
original_file_name	VARCHAR(255)	NOT NULL
stored_file_name	VARCHAR(255)	NOT NULL
file_path	VARCHAR(255)	NOT NULL
file_size	BIGINT	NOT NULL

Column Name	Data Type	Constraint
version_number	INT	DEFAULT 1
effective_date	DATE	NULL
expiry_date	DATE	NULL
access_level	ENUM('HR','Manager','Employee','All')	DEFAULT 'HR'
status	ENUM('Draft','Published','Archived')	DEFAULT 'Draft'
uploaded_by	INT	FK(Employee)
uploaded_on	DATETIME	CURRENT_TIMESTAMP

Table DOC003 : employee_document

Column Name	Data Type	Constraint
employee_document_id	INT	PK
employee_id	INT	FK(Employee)
document_type_id	INT	FK(Document Type)
document_number	VARCHAR(100)	NULL
original_file_name	VARCHAR(255)	NOT NULL
stored_file_name	VARCHAR(255)	NOT NULL
file_path	VARCHAR(255)	NOT NULL
file_size	BIGINT	NOT NULL
issue_date	DATE	NULL
expiry_date	DATE	NULL
remarks	VARCHAR(255)	NULL
verification_status	ENUM('Pending','Verified','Rejected')	DEFAULT 'Pending'

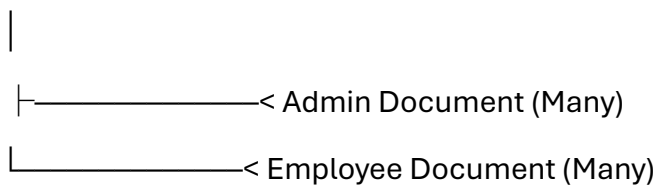
Column Name	Data Type	Constraint
verified_by	INT	FK(Employee)
verified_on	DATETIME	NULL
uploaded_on	DATETIME	CURRENT_TIMESTAMP

Table DOC004 : document_verification_log

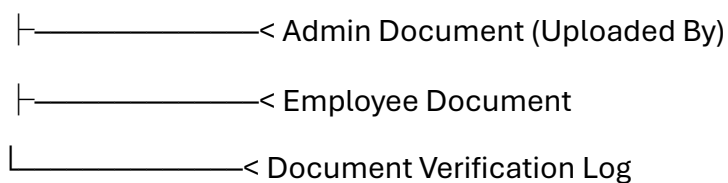
Column Name	Data Type	Constraint
verification_log_id	INT	PK
employee_document_id	INT	FK(Employee Document)
previous_status	VARCHAR(20)	NOT NULL
new_status	VARCHAR(20)	NOT NULL
remarks	VARCHAR(255)	NULL
changed_by	INT	FK(Employee)
changed_on	DATETIME	CURRENT_TIMESTAMP

Entity Relationship

Document Type (1)



Employee (1)



Dashboard Widgets

- Total Documents
- Pending Verification
- Verified Documents
- Rejected Documents
- Expiring Soon
- Expired Documents
- Mandatory Documents Missing
- Recent Uploads

Security Considerations

- Files are stored outside the public web root (recommended).
- File download requires authentication.
- Employees can access only their own documents.
- Sensitive documents are encrypted during storage (optional enhancement).
- Audit logs are maintained for verification actions.

Business Rules

- Mandatory onboarding documents must be uploaded before employee confirmation.
- Verified documents cannot be modified directly; a new version must be uploaded.
- Expired documents trigger automatic alerts.
- Document deletion is restricted for verified and published records.
- File access is controlled through role-based authorization.

Deliverables

- Document Type Master
- Admin Document Repository
- Employee Document Repository
- Secure File Upload & Download

- Document Verification Workflow
- Document Expiry Tracking
- Document Reports
- Document Dashboard
- Audit & Security Controls

11. Employee Movement Module (Promotion, Resignation & Termination)

11.1 Module Overview

The Employee Movement Module manages the complete employee lifecycle after joining the organization. It records promotions, department/designation transfers, resignations, notice periods, terminations, exit interviews, clearance processes, and final settlement.

This module integrates with Employee, Payroll, Attendance, Leave, and Document Management modules to maintain a complete employee history.

11.2 Module Objectives

The Employee Movement Module aims to:

- Manage employee promotions.
- Record department/designation transfers.
- Process employee resignations.
- Track notice periods.
- Manage employee terminations.
- Conduct exit interviews.
- Process No Due Clearance.
- Integrate with payroll for Full & Final Settlement.

- Maintain employee movement history.
-

11.3 User Roles

User Role	Permission
Administrator	Full Access
HR Manager	Full Access
HR Executive	Manage Employee Movement
Department Manager	Recommend Promotion / Approve Resignation
Employee	Submit Resignation & View Status

11.4 Module Components

The Employee Movement Module consists of:

1. Promotion Management
 2. Transfer Management
 3. Resignation Management
 4. Notice Period Management
 5. Termination Management
 6. Exit Interview
 7. No Due Clearance
 8. Full & Final Settlement
 9. Employee Movement Reports
-

11.5 Employee Movement Workflow

Employee Lifecycle



Promotion / Transfer



Resignation Submitted



Manager Approval



Notice Period



Exit Interview





No Due Clearance



Final Settlement



Employee Exit

Functional Requirements

FR-EMP-001

Promotion Management

Description

The Promotion Management module maintains employee promotion records.

Functions

- Add Promotion
- Update Promotion
- Cancel Promotion
- View Promotion History

Promotion Details

Field	Type
Promotion ID	Auto
Employee	Dropdown
Promotion Date	Date
Old Department	Auto
New Department	Dropdown
Old Designation	Auto
New Designation	Dropdown
Previous Salary	Decimal
Revised Salary	Decimal
Promotion Reason	TextArea
Approved By	Manager
Status	Pending / Approved / Rejected

Business Rules

- Promotion updates employee designation automatically.
- Salary revision is effective from Promotion Date.
- Promotion history cannot be deleted.

FR-EMP-002

Transfer Management

Description

Transfer employees between departments, branches, or reporting managers.

Functions

- Transfer Employee
 - Update Transfer
 - Cancel Transfer
 - View Transfer History
-

Transfer Details

Field	Type
Transfer ID	Auto
Employee	Dropdown
Transfer Type	Department / Branch / Manager
From Location	Text
To Location	Text
Effective Date	Date
Reason	TextArea
Approved By	Manager
Status	Pending / Approved

Business Rules

- Transfer updates employee profile automatically.
 - Historical transfers remain available for reporting.
-

FR-EMP-003

Resignation Management

Description

Employees can submit resignation requests through HRMS.

Functions

- Apply Resignation
 - Withdraw Resignation
 - Approve / Reject
 - Track Status
-

Resignation Details

Field	Type
Resignation ID	Auto
Employee	Auto
Resignation Date	Date
Last Working Date	Date
Reason	TextArea
Notice Period (Days)	Auto
Manager Remarks	TextArea
HR Remarks	TextArea
Status	Pending / Approved / Rejected / Withdrawn

Business Rules

- Last Working Date must be after Resignation Date.
- Employees cannot resign twice simultaneously.
- Withdrawal is allowed before approval.

FR-EMP-004

Notice Period Management

Description

Tracks employee notice period after resignation.

Features

- Notice Start Date
 - Notice End Date
 - Remaining Days
 - Buyout Option
 - Notice Completion
-

Notice Details

Field	Type
Notice ID	Auto
Employee	Dropdown
Notice Start Date	Date
Notice End Date	Date
Notice Period	Number
Buyout	Yes / No

Field	Type
Buyout Amount	Decimal
Status	Running / Completed

Business Rules

- Notice period is calculated automatically.
- Buyout amount is deducted during final settlement.

FR-EMP-005

Termination Management

Description

HR manages employee termination records.

Functions

- Terminate Employee
- Record Reason
- Upload Documents
- View Termination History

Termination Details

Field	Type
Termination ID	Auto
Employee	Dropdown
Termination Date	Date

Field	Type
Termination Type	Performance / Misconduct / Layoff / Contract End
Reason	TextArea
Approved By	HR Manager
Supporting Document	Upload
Status	Active / Completed

Business Rules

- Employee login is disabled after termination.
 - Payroll is stopped after termination date.
-

FR-EMP-006

Exit Interview

Description

HR conducts an exit interview before relieving the employee.

Exit Interview Details

Field	Type
Exit Interview ID	Auto
Employee	Dropdown
Interview Date	Date
Interviewer	HR
Reason for Leaving	TextArea

Field	Type
Company Rating	1-5
Manager Rating	1-5
Suggestions	TextArea

FR-EMP-007

No Due Clearance

Description

Collect department-wise clearance before employee exit.

Departments

- HR
 - IT
 - Finance
 - Admin
 - Security
-

Clearance Details

Field	Type
Clearance ID	Auto
Employee	Dropdown
Department	Dropdown
Clearance Status	Pending / Approved

Field	Type
Remarks	TextArea
Approved By	Employee

Business Rules

- Final settlement starts only after all departments approve clearance.

FR-EMP-008

Full & Final Settlement

Description

Calculate employee settlement amount after resignation or termination.

Settlement Includes

Earnings

- Pending Salary
- Leave Encashment
- Incentive
- Bonus

Deductions

- Notice Buyout
- Loan Recovery
- Asset Recovery
- Other Deduction

Reports

The Employee Movement Module provides:

- Promotion Report
- Transfer Report
- Resignation Report
- Notice Period Report
- Termination Report
- Exit Interview Report
- Clearance Report
- Final Settlement Report
- Employee Movement History

Database Design

Table EMP001 : promotion

Column Name	Data Type	Constraint
promotion_id	INT	PK
employee_id	INT	FK(Employee)
promotion_date	DATE	NOT NULL
old_department_id	INT	FK(Department)
new_department_id	INT	FK(Department)
old_designation_id	INT	FK(Designation)
new_designation_id	INT	FK(Designation)
previous_salary	DECIMAL(12,2)	NOT NULL
revised_salary	DECIMAL(12,2)	NOT NULL

Column Name	Data Type	Constraint
reason	TEXT	NULL
approved_by	INT	FK(Employee)
status	ENUM('Pending','Approved','Rejected')	DEFAULT 'Pending'

Table EMP002 : transfer

Column Name	Data Type	Constraint
transfer_id	INT	PK
employee_id	INT	FK(Employee)
transfer_type	ENUM('Department','Branch','Manager')	NOT NULL
from_location	VARCHAR(150)	NOT NULL
to_location	VARCHAR(150)	NOT NULL
effective_date	DATE	NOT NULL
reason	TEXT	NULL
approved_by	INT	FK(Employee)
status	ENUM('Pending','Approved')	DEFAULT 'Pending'

Table EMP003 : resignation

Column Name	Data Type	Constraint
resignation_id	INT	PK
employee_id	INT	FK(Employee)
resignation_date	DATE	NOT NULL

Column Name	Data Type	Constraint
last_working_date	DATE	NOT NULL
reason	TEXT	NOT NULL
notice_period_days	INT	NOT NULL
manager_remarks	TEXT	NULL
hr_remarks	TEXT	NULL
status	ENUM('Pending','Approved','Rejected','Withdrawn')	DEFAULT 'Pending'

Table EMP004 : notice_period

Column Name	Data Type	Constraint
notice_id	INT	PK
resignation_id	INT	FK(Resignation)
notice_start_date	DATE	NOT NULL
notice_end_date	DATE	NOT NULL
buyout	BOOLEAN	DEFAULT FALSE
buyout_amount	DECIMAL(12,2)	DEFAULT 0
status	ENUM('Running','Completed')	DEFAULT 'Running'

Table EMP005 : termination

Column Name	Data Type	Constraint
termination_id	INT	PK
employee_id	INT	FK(Employee)

Column Name	Data Type	Constraint
termination_date	DATE	NOT NULL
termination_type	ENUM('Performance','Misconduct','Layoff','Contract End')	NOT NULL
reason	TEXT	NOT NULL
supporting_document	VARCHAR(255)	NULL
approved_by	INT	FK(Employee)
status	ENUM('Active','Completed')	DEFAULT 'Active'

Table EMP006 : exit_interview

Column Name	Data Type	Constraint
exit_interview_id	INT	PK
employee_id	INT	FK(Employee)
interview_date	DATE	NOT NULL
interviewer_id	INT	FK(Employee)
reason_for_leaving	TEXT	NULL
company_rating	INT	CHECK (company_rating BETWEEN 1 AND 5)
manager_rating	INT	CHECK (manager_rating BETWEEN 1 AND 5)
suggestions	TEXT	NULL

Table EMP007 : no_due_clearance

Column Name	Data Type	Constraint
clearance_id	INT	PK
employee_id	INT	FK(Employee)
department_id	INT	FK(Department)
clearance_status	ENUM('Pending','Approved')	DEFAULT 'Pending'
remarks	TEXT	NULL
approved_by	INT	FK(Employee)
approved_on	DATETIME	CURRENT_TIMESTAMP

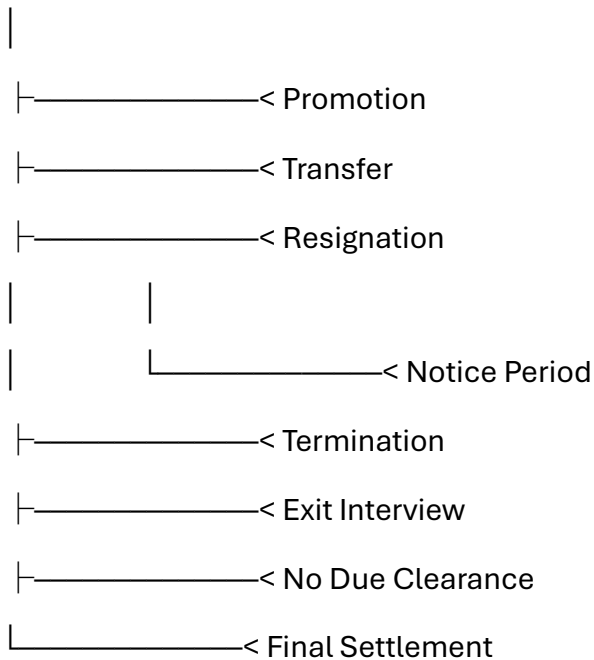
Table EMP008 : final_settlement

Column Name	Data Type	Constraint
settlement_id	INT	PK
employee_id	INT	FK(Employee)
resignation_id	INT	FK(Resignation), NULL
termination_id	INT	FK(Termination), NULL
pending_salary	DECIMAL(12,2)	DEFAULT 0
leave_encashment	DECIMAL(12,2)	DEFAULT 0
bonus	DECIMAL(12,2)	DEFAULT 0
incentive	DECIMAL(12,2)	DEFAULT 0
loan_recovery	DECIMAL(12,2)	DEFAULT 0
notice_buyout	DECIMAL(12,2)	DEFAULT 0
other_deductions	DECIMAL(12,2)	DEFAULT 0

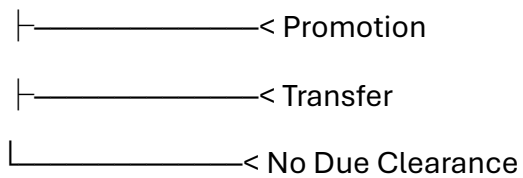
Column Name	Data Type	Constraint
net_settlement	DECIMAL(12,2)	NOT NULL
settlement_date	DATE	NOT NULL

Entity Relationship

Employee (1)



Department (1)



Dashboard Widgets

- Pending Promotions
- Pending Transfers

- Pending Resignation Approvals
 - Employees in Notice Period
 - Upcoming Last Working Day
 - Pending No Due Clearances
 - Completed Final Settlements
 - Employee Exit Summary
 - Employee Movement Trend
-

Business Rules

- Promotion updates the employee's department, designation, and salary after approval.
 - Only one active resignation request is allowed per employee.
 - Employee access is disabled automatically on the Last Working Date or Termination Date.
 - Full & Final Settlement can be processed only after all No Due Clearances are approved.
 - All employee movement records are retained permanently for audit and compliance.
-

Deliverables

- Promotion Management
- Transfer Management
- Resignation Management
- Notice Period Tracking
- Termination Management
- Exit Interview
- No Due Clearance

- Full & Final Settlement
 - Employee Movement Reports
 - Employee Movement Dashboard
-

12. Help Desk (Raise Ticket) & Reports Module

12.1 Module Overview

The Help Desk (Raise Ticket) Module enables employees to report issues, request services, and track their requests until resolution. Employees can raise tickets related to HR, IT, Admin, Payroll, Attendance, Leave, Documents, and other departments.

The Reports Module provides comprehensive MIS (Management Information System) reports and dashboards across all HRMS modules for management decision-making.

12.2 Module Objectives

The Help Desk & Reports Module aims to:

- Allow employees to raise service requests.
 - Route tickets to the appropriate department.
 - Assign tickets to support executives.
 - Track ticket lifecycle.
 - Define SLA (Service Level Agreement).
 - Escalate overdue tickets.
 - Generate ticket analytics.
 - Provide reports across all HRMS modules.
-

12.3 User Roles

User Role	Permission
Administrator	Full Access
HR Manager	Manage HR Tickets & Reports
IT Manager	Manage IT Tickets
Department Manager	Assign Tickets
Support Executive	Resolve Assigned Tickets
Employee	Raise & Track Tickets

12.4 Module Components

The Help Desk Module consists of:

1. Ticket Category Master
2. Raise Ticket
3. Ticket Assignment
4. Ticket Comments
5. Ticket Status Tracking
6. SLA Management
7. Escalation Management
8. Reports & Dashboard

12.5 Ticket Workflow

Employee

|



Raise Ticket



Assigned to Department



Support Executive



Work In Progress



Resolved

|



Employee Verification

|



Closed

Functional Requirements

FR-TKT-001

Ticket Category Master

Description

Defines all categories available for ticket creation.

Functions

- Add Category
- Update Category
- Delete Category
- Activate / Deactivate
- Search Category

Categories

- HR
- Payroll
- Attendance
- Leave
- IT Support
- Software Issue
- Hardware Issue
- Admin
- Finance
- Network
- Email
- Document Verification

Category Details

Field	Type
Category ID	Auto
Category Name	Text
Department	Dropdown
SLA Hours	Number
Status	Active

Business Rules

- Category Name must be unique.

- SLA Hours cannot be negative.

FR-TKT-002

Raise Ticket

Description

Employees can submit support requests.

Functions

- Raise Ticket
 - Edit Ticket
 - Cancel Ticket
 - View Ticket
 - Search Ticket
-

Ticket Details

Field	Type
Ticket ID	Auto
Employee	Auto
Ticket Subject	Text
Category	Dropdown
Priority	Low / Medium / High / Critical
Description	TextArea
Attachment	Upload
Raised Date	DateTime

Field	Type
Status	Open

Business Rules

- Subject is mandatory.
- Attachment is optional.
- Ticket Number is auto-generated.

FR-TKT-003

Ticket Assignment

Description

Assign tickets to the appropriate support executive.

Functions

- Assign Ticket
- Reassign Ticket
- Change Priority
- Update Status

Assignment Details

Field	Type
Assignment ID	Auto
Ticket	Dropdown
Assigned To	Employee

Field	Type
Assigned By	Manager
Assigned Date	Date
Due Date	Date
Status	Assigned

Business Rules

- One ticket can have only one active assignee.
- Reassignment history is maintained.

FR-TKT-004

Ticket Comments

Description

Communication between employee and support executive.

Functions

- Add Comment
- Reply
- Edit
- Delete

Comment Details

Field	Type
Comment ID	Auto

Field	Type
Ticket	Dropdown
Employee	Dropdown
Comment	TextArea
Attachment	Upload
Comment Date	DateTime

FR-TKT-005

Ticket Status Management

Description

Tracks complete ticket lifecycle.

Ticket Status

- Open
 - Assigned
 - In Progress
 - Waiting for User
 - Resolved
 - Closed
 - Reopened
 - Cancelled
-

Workflow

Open

↓

Assigned

↓

In Progress

↓

Resolved

↓

Employee Verification

↓

Closed

Business Rules

- Closed tickets cannot be edited.
- Employee can reopen a resolved ticket within configurable days.
- Every status change is recorded in the audit log.

FR-TKT-006

Reports Module

Description

Provides reports from all HRMS modules.

Employee Reports

- Employee List
 - Department Report
 - Designation Report
 - Joining Report
-

Attendance Reports

- Daily Attendance
 - Monthly Attendance
 - Late Arrival
 - Overtime
 - WFH Report
-

Leave Reports

- Leave Balance
 - Leave History
 - Pending Approval
 - Leave Encashment
-

Payroll Reports

- Salary Register
- Salary Slip

- Payroll Summary
 - PF Report
 - PT Report
 - TDS Report
-

Training Reports

- Trainer Report
 - Training Schedule
 - Attendance Report
 - Feedback Report
-

Project Reports

- Active Projects
 - Project Progress
 - Overdue Tasks
 - Team Allocation
-

Ticket Reports

- Open Tickets
 - Closed Tickets
 - Pending Tickets
 - SLA Report
 - Department-wise Tickets
 - Employee-wise Tickets
-

HR Analytics Reports

- Employee Turnover
 - Attrition Rate
 - Promotion History
 - Resignation Report
 - Termination Report
-

Dashboard

The HRMS Dashboard provides:

Employee Summary

- Total Employees
 - Active Employees
 - New Joiners
 - Employees on Leave
-

Attendance Summary

- Present
 - Absent
 - Late Employees
 - WFH Employees
-

Payroll Summary

- Payroll Processed
 - Pending Payroll
 - Gross Salary
 - Net Salary
-

Leave Summary

- Pending Leaves
 - Approved Leaves
 - Leave Balance
-

Project Summary

- Active Projects
 - Completed Projects
 - Pending Tasks
-

Ticket Summary

- Open Tickets
 - Assigned Tickets
 - Resolved Tickets
 - SLA Breached Tickets
-

HR Analytics

- Attrition Rate
 - Department-wise Employees
 - Employee Growth
 - Monthly Hiring
 - Monthly Exit
 - Salary Trend
-

Database Design

Table TKT001 : ticket_category

Column Name	Data Type	Constraint
category_id	INT	PK
category_name	VARCHAR(100)	UNIQUE
department_id	INT	FK(Department)
sla_hours	INT	NOT NULL
status	ENUM('Active','Inactive')	DEFAULT 'Active'

Table TKT002 : ticket

Column Name	Data Type	Constraint
ticket_id	INT	PK
ticket_number	VARCHAR(30)	UNIQUE
employee_id	INT	FK(Employee)
category_id	INT	FK(Ticket Category)
subject	VARCHAR(200)	NOT NULL
description	TEXT	NOT NULL
priority	ENUM('Low','Medium','High','Critical')	DEFAULT 'Medium'
attachment	VARCHAR(255)	NULL
raised_date	DATETIME	CURRENT_TIMESTAMP
status	ENUM('Open','Assigned','In Progress','Waiting for User','Resolved','Closed','Reopened','Cancelled')	DEFAULT 'Open'

Table TKT003 : ticket_assignment

Column Name	Data Type	Constraint
assignment_id	INT	PK
ticket_id	INT	FK(Ticket)
assigned_to	INT	FK(Employee)
assigned_by	INT	FK(Employee)
assigned_date	DATETIME	CURRENT_TIMESTAMP
due_date	DATETIME	NOT NULL
status	ENUM('Assigned','Reassigned')	DEFAULT 'Assigned'

Table TKT004 : ticket_comment

Column Name	Data Type	Constraint
comment_id	INT	PK
ticket_id	INT	FK(Ticket)
employee_id	INT	FK(Employee)
comment	TEXT	NOT NULL
attachment	VARCHAR(255)	NULL
comment_date	DATETIME	CURRENT_TIMESTAMP

Table TKT005 : ticket_history

Column Name	Data Type	Constraint
history_id	INT	PK
ticket_id	INT	FK(Ticket)

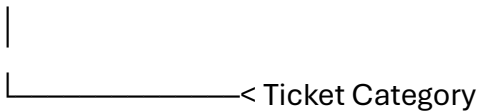
Column Name	Data Type	Constraint
old_status	VARCHAR(50)	NOT NULL
new_status	VARCHAR(50)	NOT NULL
changed_by	INT	FK(Employee)
changed_on	DATETIME	CURRENT_TIMESTAMP
remarks	VARCHAR(255)	NULL

Table TKT006 : ticket_escalation

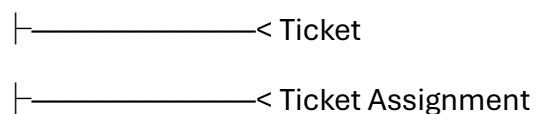
Column Name	Data Type	Constraint
escalation_id	INT	PK
ticket_id	INT	FK(Ticket)
escalation_level	INT	NOT NULL
escalated_to	INT	FK(Employee)
escalation_date	DATETIME	CURRENT_TIMESTAMP
reason	VARCHAR(255)	NULL

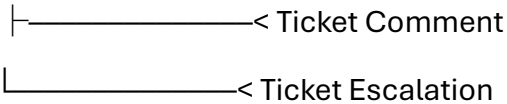
Entity Relationship

Department (1)

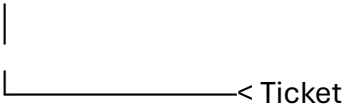


Employee (1)

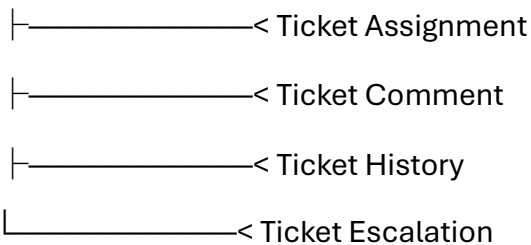




Ticket Category (1)



Ticket (1)



Business Rules

- Every ticket must belong to one category.
- Ticket Number is generated automatically (e.g., **TKT-20260730-0001**).
- SLA countdown begins immediately after ticket creation.
- Employees can reopen resolved tickets within the configured period.
- All ticket actions (assignment, comments, status changes, escalations) are stored in the audit history.
- Closed tickets are read-only and available for reporting.

Dashboard Widgets

- Total Tickets
- Open Tickets
- Assigned Tickets

- In Progress Tickets
 - Resolved Tickets
 - Closed Tickets
 - SLA Breached Tickets
 - Average Resolution Time
 - Department-wise Ticket Count
 - Employee-wise Ticket Summary
-

Deliverables

- Ticket Category Master
 - Raise Ticket Module
 - Ticket Assignment
 - Ticket Status Workflow
 - Ticket Comments
 - SLA Management
 - Escalation Management
 - Reports & Analytics Dashboard
 - Ticket Reports
 - Audit Trail
-

HRMS Project Conclusion

The HRMS application consists of the following integrated modules:

1. Employee Management
2. Event & Holiday Management
3. Training Management
4. Attendance & Timesheet Management

5. Leave Management
6. Payroll Management
7. Project Management
8. Document Management
9. Employee Movement (Promotion, Transfer, Resignation, Termination)
10. Help Desk (Raise Ticket)
11. Reports & Analytics Dashboard